Hero Companion

Help & Release Notes — your City of Heroes sidekick

App version 0.12.58 · Model v56 · Game database 2026.1.1242
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Hero Companion is a fan-made build planner for City of Heroes: Homecoming. It plans, optimizes, and walks you through character builds using the game's actual math — to-hit, resistance, defense, endurance, recharge — not templates or guesswork. It runs entirely on your own computer, next to (never inside) the game.

Quick Start

Open the app and pick one of the entry cards:

- **Build a new level 50** — you choose archetype, powersets, the content you'll run, and your role on a team; the optimizer designs a complete end-game build with enhancement slotting.
- **Start from scratch (level 1)** — same choices, but you get a level-by-level leveling companion that walks every pick and slot from 1 to 50.
- **Import from Mids Reborn** — paste a Mids build (or load a .mbd file) and the tool evaluates it, then can optimize from there.
- **Import a character you play** — type `/build_save_file` in game, then click "Find my characters for me": the tool locates your Homecoming saves itself and imports the one you pick. (Unusual install location? Tell it your game folder once and it remembers.)

Everything is reversible. Nothing you click is permanent, and in-game a /respec rewrites any character anyway.

The Build Panel

Pick your archetype, primary, and secondary. The panel fills with your powers; click a power to add or remove it, and use the slot controls to place enhancement slots (the game grants 67 placeable slots across a career; the tool enforces the real limits).

The stat panel updates live: defense by position, resistance by type, recharge, recovery, damage. Caps are shown — defense soft cap 45%, resistance cap by archetype.

Reading the Numbers

Every stat is drawn as a bar, and each bar tells you three things at once.

- **The filled portion** is what your build actually has right now.
- **The marked line** is the number that matters for that stat: 45% for defense (the soft cap, where most incoming attacks start missing), or your archetype's resistance cap, which is

90% for Tankers and Brutes, 85% for Kheldians and the Arachnos archetypes, and 75% for everyone else. Past that line the game stops counting, so points spent beyond it are wasted.

- **The number itself**, so you never have to estimate from the picture.

Two things on the bars are worth knowing about, because they surprise people:

" N%" on a defense row. Some defense (anything from a Hide-style power) switches off the moment you attack. When that applies, the row shows both numbers: the resting one, and the in-combat one after the drop. The in-combat figure is the one to plan around, because that is what you have when it matters.

Achieved vs target. After a solve you get a list comparing what you asked for against what the build reached. A row that fell short is not the tool failing quietly — it will also tell you which unpicked powers on your character would help and roughly how much each would add. If nothing on your character supplies that stat, it says so plainly, because a goal you cannot reach is worth knowing about rather than chasing.

How pet damage is counted. Pets and henchmen are scored with their real chance to hit, at their own level — not yours. The game summons lower-tier henchmen below your level (at full strength, tier 1 fights two levels down, tier 2 one, tier 3 at your level), so against high-level enemies they miss more, and the tool counts that honestly instead of pretending every pet always hits. It also counts what buys the accuracy back: ToHit buffs that reach your pets (the Mastermind's own Supremacy, Tactics) and accuracy slotted in the summon power itself. That is why those choices visibly raise pet damage here, just as they do in the game.

Hamidon Origins. A Hamidon Origin enhancement carries two or three aspects in one slot — accuracy and damage together, or resistance and endurance — which is more raw enhancement per slot than any set piece. The price is that HOs earn no set bonuses, and they come from endgame play: Hamidon raids, or merit conversion. The optimizer may propose them where they genuinely beat a set piece — mostly for endgame content (incarnate trials and farms), and as the accuracy core of proc-heavy attacks — and every HO it places says where it comes from. **Don't have HOs?** Untick *I have Hamidon Origins (HOs) to slot* (under *Customize build targets…*) and no build will slot one. You can still slot them by hand anywhere, for any content.

Content and Role

These two dropdowns are the most important choice in the tool:

- **Content** — what you actually run: general play, task forces, incarnate trials, fire farms, PvP. It sets what the build must survive and deliver.
- **Role** — what you're there to do: damage, tank, buff/debuff support, control, healing. The optimizer maximizes your role output, not a generic score.

A support character is scored on how much its debuffs and buffs actually change fights (magnitude times uptime), not on its own survival. That's deliberate: the tool builds characters that are noticed as contributors.

The Optimizer

The "Optimize" actions run a real search over powers, pools, and slotting:

- It explores every legal add, drop, and swap — including all power pools — and runs until no single change improves the build (with honest certificates saying so).
- It respects the game's rules: pool limits (four max), the origin-pool rule (only one of Sorcery / Experimentation / Force of Will / Gadgetry / Utility Belt), archetype set access, prerequisite tiers, and level availability.
- It learns: strong past results seed future searches as champions.

If a suggestion looks odd, it earned its place in the math — but you always have the final say. Swap anything; the tool re-evaluates around your choices.

Locking a Power

Every power card has a padlock: open means the optimizer may change it, closed means hands off.

A lock is absolute. A locked power comes out of a re-solve **exactly** as you left it, down to the individual enhancements. Everything unlocked is re-slotted toward your goal around it.

Weak attacks carrying sets. The optimizer only counts an attack's damage if the attack chain would actually use it. A low-tier attack (Boxing, Kick, or a set's first attacks) can still carry a set when its set bonuses are worth the slots — the card says so, and the bonuses work whether or not you use the attack. Prefer to keep it light? Slot it how you like and close its padlock.

Use it when you have already decided something and want the tool to work with that decision rather than argue with it. Common cases: you have expensive sets already slotted in game and do not want a plan that assumes you will re-buy them; you have a favourite proc in a particular power; or you are exploring "what would change if this part stayed fixed".

Two consequences worth knowing before you lock:

- **An empty slot inside a locked power stays empty.** The lock means untouched, not "untouched except the gaps". The tool will tell you when that has happened rather than quietly leaving holes.
- **Locking constrains the result.** A locked power that is slotted poorly limits how good the rest can be, and the solve is being honest when it comes back lower. If a target suddenly cannot be reached, your locks are the first thing to check.

Locks are saved with the character, so resuming later finds everything exactly as you left it.

If you want most of a build kept rather than specific powers, use **Preserve my IO sets** instead: that keeps the sets you have already invested in and only re-slots generic enhancements and empty slots. Locks are the scalpel; preserve is the broad stroke.

Leveling from 1 to 50

For from-scratch characters, the leveling walk shows exactly what happens at every level on the real

Homecoming schedule: a power on even levels, slots on odd levels, pools at 4, epics at 35, respec trials at 24/34/44, and cost-smart enhancement advice for each stretch (cheap early, common IOs from 7, sets near the end).

You can take every suggestion or none of them. The walk tracks where you've deviated and offers to re-fit the end-game around your actual choices.

Kheldians (Peacebringer / Warshade)

Kheldians follow their own rules and the tool knows it: inherent flight from level 1 (no travel pool needed unless you want one), Nova and Dwarf forms inside your own power sets, and no epic pool at all — the walk and the wizard both reflect that.

Arachnos Soldiers and Widows

VEATs live a two-phase career and the walk follows it honestly:

- **Levels 1–23**: only your base sets (Arachnos Soldier / Training and Gadgets, or Widow Training / Teamwork) — branch powers can't be taken yet, so the walk never suggests them.
- **Level 24**: the mandatory respec. You choose your branch — Crab or Bane, Night Widow or Fortunata — and re-place every pick from level 1 with all six sets open. The walk hands you the complete re-place order right at this step.
- **Levels 24–50**: the walk continues from the respec order, branch powers included.

Enhancement Converter and Haul Appraiser

The Converter panel answers two questions:

- **"How do I get this IO cheaply?"** — a concrete cheapest path per enhancement: which piece to buy or craft, which converters to use, and the converter/merit cost. One rule of the game worth knowing: conversions can't jump from cheap pools into purples — the tool's paths never pretend otherwise.
- **"Is this drop worth anything?"** — paste your drops (recipes included, straight from the game). Each item gets a verdict: keep for your build, craft-then-convert for profit, or just sell.

Saving, Importing, Exporting

- **Save** keeps the character's plan and leveling progress locally so you can resume. Auto-save runs in the background. A saved character is Hero Companion's own file — Mids can't open it; use Export for that.
- **Import/Export** is Mids Reborn compatible — bring builds in, take builds out. **Export to Mids Reborn** writes a `.mbd` file (open it in Mids with **File → Open**) and copies a Mids build code to your clipboard (paste it in Mids under **Build Sharing → Import DataChunk**). Your builds are yours; nothing leaves your machine.

Your alignment

The alignment button in the banner shows your current alignment. Click it and you get all four: **Hero**, **Vigilante**, **Rogue**, **Villain**. It reskins the whole app, and your choice is remembered.

The two middle alignments are not a third side. A Vigilante levels in Paragon City like a hero and can also visit villain content; a Rogue levels in the Rogue Isles like a villain and can also visit hero content. That is how the game works, and it is why they are shown in gold rather than blue or red.

Your build scores the same either way. The game counts a Vigilante as hero-side and a Rogue as villain-side, so Hero and Vigilante give identical numbers, and so do Villain and Rogue. Changing alignment changes what you *see*, never what your build is worth.

Moving between the hero pair and the villain pair can shift your totals slightly. Accolades are alignment-specific in game, and the app assumes the standard set for your side, so a villain build should not be counting Freedom Phalanx Reserve. Accolades carry real stats, so the numbers move a little. That is correct behaviour, not a glitch.

Nothing you ticked is ever thrown away. An accolade for the other side stays remembered, greys out, and counts as zero while you are on this side — the game gates it the same way, so a held off-side accolade does nothing until you switch back. Switch back and it counts again. Powers, slotting and targets are never touched by any of this.

A separate control, easy to confuse with this one: the Leveling Guide has its own alignment switch, which additionally offers Flashback. That one is a **preview** of somebody else's route through the game, so you can see where a villain would level. It changes nothing about your character and resets when you leav e the Leveling Guide.

Bugs, Champions, and Updates

Hero Companion is a living tool, and you're part of how it improves. Three buttons, all strictly opt-in — the app never sends anything on its own:

- **Report a bug** — opens a pre-filled bug report on the project's GitHub page with your app, model, and game-data versions already included (that's usually the hard part of a good bug report). Add what happened and post it. Nothing is sent until you click submit on GitHub itself.
- **Submit champion** — think your current build beats the shipped champion for its archetype and role? This saves your build as a *champion candidate* file and opens the submission queue. The development hub re-scores every candidate with its own math — if your build genuinely wins, it becomes the new shipped champion in a future update, with credit to you.
- **check for updates** (bottom of the page) — compares your version against the latest release on GitHub. On first run the app asks once whether to do this check automatically at startup (it contacts github.com to compare version numbers and sends nothing else); say yes and new releases greet you with an "Update now / Remind me later" banner. Say no and the footer button is the only check that ever runs. Updates are never downloaded or installed for you.

A few words the tool uses precisely: a **champion** is the best certified build for an archetype + powersets + role combination — champions ship with the app and give the optimizer its head start. A **candidate** is your claim to beat one. The **model version** stamps which generation of the scoring physics produced a number, and the **data version** stamps which game database it was computed against — so scores are always compared apples to apples.

Frequently Asked

The tool and the game disagree — who's right? The game, always. The tool's model is verified against the Homecoming Wiki and City of Data, but patches happen. When you find a disagreement, that's a bug worth reporting.

Why won't it give me a second origin pool / a branch power at level 10 / an epic on my Peacebringer? Because the game won't allow it. The tool only suggests choices that exist for your character at that level.

Does it play the game for me? No. It never touches the game client or servers. It's a planner you read while you play.

Is it free? Free and noncommercial, forever (CC BY-NC-SA 4.0). See the Terms of Use, License, and Credits links at the bottom of the app.

Credits

Built on the work of the City of Heroes community: NCSoft (the game, all rights theirs), Homecoming Servers — whose game client this tool now reads its data directly from — Mids Reborn (the build planner it was built on, and stays `.mbd` import/export compatible with), the Unofficial Homecoming Wiki, City of Data v2, the Homecoming Forums, the Paragon Wiki Archive — Guyver [SoV] of the Sovereign supergroup, whose openly shared master builds were the standard this tool's optimizer had to honestly beat — and Maelwys, whose expert forum reviews caught what our own tests missed, round after round.

Hero Companion — What's New

0.12.58 — 2026-10-13 — Detonator, per the beta tests

Two more Detonator answers from Maelwys (forum topic 64761), both tested on the beta servers.

- **An armed Acid Mortar blows when it runs out.** When its 60 seconds end, a Mortar carrying a Detonator bomb explodes, the same as if it were defeated. The app now counts that blast: once per Detonator cast, or once per Mortar summon if you can't resummon it that often.

- **The bomb uses both powers' slotting.** Damage and Accuracy slotted in the pet's summon power and in Detonator now both reach the blast. They're added together first and then hit the usual diminishing returns, the same way a Mastermind upgrade passes slotting to henchmen. Before, the app only used Detonator's slotting.

- **A Force Field Generator's bomb now counts when it's defeated.** It's the last Detonator carrier the app wasn't scoring. It now gets the same death math as henchmen, using its own game numbers: its hit points, the defense from its own bubble (raised by any Defense slotted in the power), and its 20% resistance to everything. It doesn't inherit your set bonuses or get Supremacy, since the game gives those only to henchmen. The Mark and Command of the Mastermind auras do reach it.

0.12.57 — 2026-10-12 — Traps and Mastermind pets, per the client

Field report from Maelwys (forum topic 64761), checked against the game client.

- **Mastermind pet auras don't stack.** Supremacy's aura on your henchmen and the Mark of Supremacy / Command of the Mastermind auras share one stacking rule in the game: your henchmen get whichever is strongest, not the total. Before, the app added them, so henchmen looked sturdier than they are.

- **Detonator, corrected.** Detonator puts a bomb on the pets near you when you cast it. A pet summoned later has no bomb until you cast Detonator again, so each cast now counts at most one blast per pet. It works on the pets the game tags for it: your henchmen (but not the Necromancy ghosts or Living Hellfire), plus Force Field Generator, Acid Mortar and Maintenance Drone. New option: "I dismiss my pets to set off Detonator". When it is ticked, every armed pet blows once per Detonator cast, and the app subtracts the time spent resummoning.

- **Trip Mine is an attack.** Since Page 4 it has no interruption, uses your archetype's damage numbers, and its damage procs fire. It now counts as an area attack you place at your feet, with the game's damage (its fire and lethal amounts were wrong before).

- **Seeker Drones count.** Each cast makes two drones that each blow once: damage, -ToHit and -Damage on what they hit, and each drone rolls your damage procs. They used to be worth nothing in the app.

- **More Page 4 pet data:** Force Field Generator stays until destroyed, and Caltrops use your archetype's damage numbers.

0.12.56 — 2026-10-11 — Nothing left unmodeled

- **ATO procs that summon a one-shot pet now count.** Defender's Bastion (heal), Scourging Blast (heal and endurance) and Vigilant Assault (absorb shield), regular and superior, are valued at the proc rate the game uses, for you and for the allies the pet reaches.

- **Cages on teammates.** New option: "I cast Sonic Cage / Refraction Shield / Detention Field on teammates". When it is ticked, these powers are valued as the ally shield (15 seconds of defense and resistance, 90 second recharge) instead of a cage on enemies.

- **Team endurance counts.** Recovery you give allies (Speed Boost, Accelerate Metabolism, Chrono Shift, Heat Loss, Painbringer, the Ageless destiny and others) is now valued. It is worth the output your teammates get back when they would otherwise run low on endurance; the teammates are the app's own recommended builds.

- **Traps Detonator.** Page 4 made every henchman explode when it dies. That blast is now valued at the rate henchmen die and can be resummoned.

- **More Page 4 corrections from the game client:** Char (Fire Mastery for Brutes, Tankers and Sentinels) now has Page 4's higher damage; PvP mode now has Page 4's PvP-only changes; Time Bomb's fire and lethal damage were swapped.

- **Recommended builds re-checked** for every i-trial build in the app.

0.12.55 — 2026-10-09 — Page 4, finished

- **Prismatic Shield now counts.** Light Affinity's Prismatic Shield places a field that, for 45 seconds, gives you and your allies defense and resistance to all damage, and makes enemies inside it deal less damage. The app gave it no credit in 0.12.54; it is now valued for the time the field is up (45 seconds out of every 150 before recharge), with the numbers read from the game client.

- **Mastermind pet auras count.** Page 4 gave Supremacy a pet aura (+10% AoE defense, +10% resistance, +100% regeneration for your henchmen). It now counts toward how long your henchmen stay up, together with the Mastermind ATO auras: Mark of Supremacy (+12.5% resistance and +125% regeneration; superior 15% / 150%) and Command of the Mastermind (+12.5% AoE defense; superior 15%).

- **Dominator ATO procs count.** The Fiery Orb (Dominating Grasp) and Energy Font (Overpowering Presence) pets are now valued with Page 4's numbers: 2 procs per minute (3 for the superior version), each pet lasts 25 seconds, at most 3 at once.

- **Fixes to 0.12.54's Page 4 data.** Checking the above turned up mistakes in how 0.12.54 read the update, now corrected from the game client: - AoE defense was missing from

Protective Beam, Prismatic Shield and Sonic Aura's armor toggle, and was left at its old value on Fortitude, Personal Force Field and Dispersion Bubble. - Web Grenade's slow, and Caltrops' new shrapnel damage, were missing. - Sonic Cage, Refraction Shield and Detention Field had lost their cage on enemies.

- **Recommended builds re-checked** for every archetype these changes touch: all Dominators and Masterminds, every Light Affinity and Sonic Aura build, and every build that uses one of the corrected powers.

- **Not modeled yet:** using Sonic Cage, Refraction Shield or Detention Field on a teammate (Page 4 added this; it changes the power's own recharge), and the three ATO procs that summon a one-shot pet (Defender's Bastion, Scourging Blast, Vigilant Assault).

- **Data currency:** Homecoming client as of Issue 28 Page 4 (October 6, 2026).

0.12.54 — 2026-10-09 — Issue 28 Page 4 is in

- **Light Affinity and Sonic Aura are here.** Light Affinity (Defender primary; Controller, Corruptor and Mastermind secondary) and Sonic Aura (Tanker primary; Brute, Scrapper, Stalker and Sentinel secondary) can now be picked, built, slotted and scored, with numbers read straight from the updated game client. 151 new recommended builds cover every archetype that can take them.

- **Page 4's power changes are in.** 289 changed powers were updated from the game client, including the Super Strength, Empathy, Force Field, Traps and Sonic Resonance reworks (Rage no longer crashes, Hand Clap now does damage, Fortitude and Regeneration Aura changed, Force Bolt hits harder, and more), plus the cast-time and endurance trims across many other sets.

- **Higher damage cap for eight archetypes.** Defenders, Controllers, Dominators, Masterminds, Arachnos Soldiers and Widows, Peacebringers and Warshades now cap at +400% damage (was +300%), as in the game.

- **Brute base hit points corrected to 1,606** (was 1,499) to match the game.
- **Recommended builds re-checked.** Every recommended build these changes could touch was re-optimized — 1,836 in all. 1,610 improved and were replaced; the other 68 were already the better build and stayed. Every one of the 2,865 recommended builds was confirmed to load.

- **Not modeled yet:** Prismatic Shield's pulsing aura, the Dominator ATO proc-rate changes and the Mastermind ATO pet-aura changes. They don't change any numbers the app shows today.

- **Data currency:** Homecoming client as of Issue 28 Page 4 (October 6, 2026).

0.12.53 — 2026-10-05 — Every unique counted, and no bad advice

- **Psionic resistance now matches Mids and the game.** Three unique enhancements were giving you nothing in the app's totals: Impervium Armor: Psionic Resistance (+6%

psionic resistance), Aegis: Psionic/Status Resistance (+5% psionic resistance) and Unbreakable Guard: +Max HP (+7.5% max HP — the power card already named it, but the totals left it out). All three now count, with the numbers taken from the game's own enhancement text. A field report showed a Stone Armor Brute at 5% psionic resistance where Mids and the game showed 11% — it now reads 11%. (Thank you for the build file!)

- **No more "take Combat Jumping" when you already have it.** An import tip suggested adding Combat Jumping as a Luck of the Gambler mule even when the build already had Combat Jumping — with the Luck of the Gambler in it. It now stays quiet in that case, and when it does show, it gives the right number: one Luck of the Gambler global is +7.5% recharge, not ~14%.

- **Fixes a step back in 0.12.52.** 0.12.52's "damage only counts where you attack" solve was always used, even on builds it made worse — measured on a Super Strength / Shield Brute, it lost a six-piece set's defense. It now competes with the normal solve, and the app's build scorer keeps whichever build is actually better. Across ten archetypes tested, no build scores lower than in 0.12.51 or 0.12.52.

- **Data currency:** unchanged (Homecoming client as of the July 7, 2026 game update — still the newest live patch; liveness clean, 0 diffs).

0.12.52 — 2026-10-02 — Damage only counts where you attack

- **The optimizer no longer credits damage to attacks you'd never use.** It used to value every attack's damage by how hard it hits — so a quick, weak attack like Jab still earned "damage" credit for its slots, even though the app's own attack-chain model (built from the game's cast times, recharge and damage) never fires it. Enhancing an attack you don't activate adds no damage in the game, so that credit is gone: a single-target attack the chain never uses has to earn its slots through set bonuses alone. Area attacks are unaffected — the app already counts them as fired whenever they recharge. (Thanks to the forum report that called this out.)

- **Power cards say why a low-tier attack is slotted.** When Boxing, Kick, or one of a set's first two attacks carries a set, its note now opens with "*Slotted for the set bonuses*" and points out that the bonuses count whether or not you use the attack. You may still see a set there: the set bonuses are often genuinely worth the slots. If you'd rather keep an attack light, close its padlock and re-solve.

- **Measured:** on the reported Super Strength / Shield Brute the build's overall score rose slightly (more survivability, a little less personal damage); four other archetype checks came out identical. Certified champion builds are unchanged.

- **Data currency:** unchanged (Homecoming client as of the July 7, 2026 game update — still the newest live patch; liveness clean, 0 diffs).

0.12.51 — 2026-10-02 — No HOs, and a Mids code you can paste

- **New: "I have Hamidon Origins (HOs) to slot."** It sits under *Customize build targets…* and is ticked by default. Untick it if you don't raid Hamidon and don't plan to buy HOs, and no build will slot one again — not in the optimizer, not in proc-heavy attacks. Attacks that used HOs for their accuracy keep pieces of their own sets instead. The app remembers your choice. (Thanks to the forum report that asked for this — it also showed HOs were turning up outside endgame content, which this stops too.)

- **Export to Mids Reborn now also copies a Mids build code.** Mids has two ways in, and only one took our file: **File Open** loads the exported .mbd, but Mids' import boxes reject it with a format error. Export now also puts a Mids build code on your clipboard — paste it into Mids under **Build Sharing Import DataChunk**. Both routes were tested in Mids Reborn 3.8.6 with its current Homecoming database.

- **Data currency:** unchanged (Homecoming client as of the July 7, 2026 game update — still the newest live patch; liveness clean, 0 diffs).

0.12.50 — 2026-09-19 — The app says who it is

- **The app now identifies itself in Windows.** Right-click the exe Properties Details was completely blank: no version, no publisher, no description. It now carries all three, and so does the installer, whose file-version field had the same gap. Nothing about how the app runs changes — but a signed program that refuses to say what it is reads as anonymous to both people and antivirus heuristics.

- **Data currency:** unchanged (Homecoming client as of the July 7, 2026 game update — still the newest live patch; liveness clean, 0 diffs).

0.12.49 — 2026-09-19 — Attuned enhancements import as what they are

- **Attuned set pieces no longer import as empty slots.** A field report (thank you!) showed a build whose Fire Sword came back with its last two slots blank — an attuned Touch of Death proc and an attuned Achilles' Heel: Chance for Res Debuff — even though both were slotted in game. The game names an attuned copy of a set differently from the crafted one, and our enhancement catalog only carried the crafted name for those sets, so the importer found nothing and left the slot empty rather than guessing. Both importers now recognise the two names as the same piece: the in-game export and the Mids .mbd file. Superior versions of Archetype sets are deliberately still treated as their own sets, because they are.

- **Imported attuned pieces are now valued at your character's level.** The game writes an attuned enhancement as level 1, because an attuned piece has no fixed level — it scales with you. Recovering the slot was only half the job; it also had to be priced right, so an attuned piece imported this way is now worth what its crafted level-50 copy would be instead of what a level-1 enhancement would. Your set bonuses and totals should match what the game shows you.

- **Data currency:** unchanged (Homecoming client as of the July 7, 2026

game update — still the newest live patch; liveness clean, 0 diffs).

0.12.48 — 2026-09-02 — The launch error, fixed at the root; forms seat where the game grants them

- **0.12.47 could fail to launch after an update — fixed, and the whole failure class with it.** Upgrading never deleted obsolete files from the app folder, so leftovers from older installs accumulated; a stray solver library fragment from a build made on the old development machine survived every upgrade and finally crashed the engine at startup. The installer now wipes the app's internal folder before laying down the new version, so an installed copy is always exactly the released build. Your saves and settings are untouched — they live in your user profile, which the installer never clears. If 0.12.47 wouldn't start for you, installing 0.12.48 over it is the fix; no uninstall needed.

- **Kheldian forms now seat at the level the game grants them** — Nova at 4, Dwarf at 20 (the known issue stated in the 0.12.47 notes, and before that a field report — thank you!). Two seating paths queued your chosen form behind every level-1 power; a chosen form is the character, so it now takes the first seat the game allows and the rest of the build plans around it. All other archetypes seat exactly as before.

- Everything from 0.12.47 rides along: the native app window restored (with a build gate so a windowless build can never ship again) and the roster of 2,714 champions — every archetype × primary × secondary combination.

- **Data currency:** unchanged from 0.12.47 (Homecoming client as of the July 7, 2026 game update; liveness clean).

0.12.47 — 2026-09-02 — The app window returns; a champion for every combo

- **Hero Companion opens in its own window again.** A field report (thank you!) caught 0.12.46 opening as a browser tab at localhost:5000 instead of the app window it had before. The cause was ours: 0.12.46 was the first release built on a new machine, and the native-window component was missing from that build environment — the app silently fell back to its browser-tab escape hatch. The window stack is restored, and the build now refuses to produce a release at all if that component is ever missing again, so this failure can't ship silently twice.

- **Every powerset combination now has a champion.** The roster grew from 29 certified contexts to 2,714: an optimization campaign ran every archetype × primary × secondary combination (2,703 builds) through the full certified pipeline. Whatever you play, autopick now serves a converged, certificate-carrying build instead of falling back to the bare heuristic picker. The original 29 deeply-certified champions are unchanged and five of them improved this round under the canonical gate (Crab Spider Soldier, Plant/Poison, Poison/Sonic, and Radiation/Radiation — twice). The new champions carry solver certificates today; the deeper canonical-score audit runs as a follow-up wave.

- **Known issue, honestly stated:** the same field report notes Kheldian form powers seat at levels 6–8 (Nova) and 38–41 (Dwarf) rather than the game's earliest 4 and 20 that the 0.12.46 notes promised. Confirmed — the leveling scheduler places forms by build priority, not earliest availability. A proper fix is queued rather than rushed into this release.

- **Data currency:** built against the Homecoming client as of the July 7, 2026 game update (latest patch at release time); liveness check clean (10,984 powers, 458 offered sets, 0 diffs).

0.12.46 — 2026-08-30 — Kheldian form choices honored

- **Kheldian form choices are now honored on every content lane.** A returning player reported (thank you!) that picking tri-form with general/solo content produced no tri-form build: certified form champions cover iTrials today, and off that lane the planner fell back to its generic picker — which never read the form answer at all. Now the picker itself honors the choice everywhere: the form power(s) you chose are locked into the picks and placed as early as the game allows (Nova at 4, Dwarf at 20), forms you declined are never taken, and human means human. When your content lane has no certified form champion yet, the wizard and the result both say so plainly — the form powers are guaranteed in, and the build reads as a strong start, never as a certified champion. The "why this question" explainer also states which route — the certified champion or the honest heuristic — will serve your exact answers.

- **Champion update: Night Widow (iTrial) improved 8.8%.** A rescued certification run from the August wave beat the shipping Night Widow / Teamwork champion under the fresh canonical check (1410.6 1535.1) and supersedes it. Eleven other rescued candidates were checked the same way and lost or tied — the incumbents stand.

- **Data currency:** built against the Homecoming client as of the July 7, 2026 game update (latest patch at release time); last full client re-export 2026-08-19.

0.12.45 — 2026-08-17 — AFK certification reads the build

- **AFK sustain now shapes the farm builds themselves (model v49).** With the honest gate in place, the AFK requirement joined the optimizer's objective: away-from-keyboard survival counts only what runs itself (regeneration, toggles, and the one auto-fire power — click heals no longer count there), judged over a ten-minute AFK stint instead of a 60-second window. All three AFK farm builds re-converged under the new physics and improved sharply (Spines/Fiery Aura canonical 45.5 499.1; Titan Weapons/Bio 15.6 292.6; Fiery Aura/Fiery Melee 25.1 202.7), now favoring capped Fire resistance and near-softcap Fire defense. The honest bottom line stays stated on every label: none of the three sustains true AFK play at any shift — the labels say exactly what each build takes and holds. Other content types are unchanged.

- **AFK farm tiers now judge the build in front of them (model v48).** The AFK certification ladder was one fixed requirement (37 HP/s at +4x8) that silently assumed capped Fire resistance and softcapped defense — it never read the build's own numbers, so a 63% Fire-resistance

build could claim a higher AFK tier than two res-capped builds with similar regeneration. Found by Maelwys (thank you again). The requirement is now computed from the build's own Fire resistance and defense using the same arithmetic the scoring model already uses, and the label states the numbers it judged against. Under the honest gate, **no current farm build sustains AFK play at any shift** — all three AFK labels now read "active play only", including the Titan Weapons / Bio Armor build previously labeled +4x8 passive. That claim is withdrawn. Scores are unchanged (this fixes the label, not the scoring).

- **The certificate's heal ledger says what it counted.** Click heals were listed for information but never summed into sustain; each row now carries an explicit counted flag so the ledger cannot be read as crediting a heal no absent player can click.

0.12.44 — 2026-08-17 — Fire farms judged as fire farms

- **Farm scenarios now deal fire damage.** The survival math used to judge every scenario with general-content damage types, so a farmer capped at 90% Fire resistance — the entire community farming doctrine — measured as fragile in its own farm. Farm certification now reads Fire resistance and Fire defense, the way the game actually plays there. Nothing outside the farm scenarios changes.
- **The AFK auto-fire pick thinks at spawn scale.** Choosing the auto-fire power by single-target numbers picked Boxing over Burn; it now weighs a whole spawn, and Burn wins, as every real farmer knows.
- **The three AFK farm builds re-certified under the corrected model:** Spines / Fiery Aura improves sharply (and keeps its honest +0x8 passive label), Fiery Aura / Fiery Melee reaches +2x8 passive, and Titan Weapons / Bio Armor now ships a build that sustains **+4x8 fully passively** (39.1 HP/s) — chosen deliberately over a build that scored 0.7% higher without the tier. Thanks once more to Maelwys, whose example builds exposed both model gaps.

0.12.43 — 2026-08-17 — Honest AFK labels, in the app

- **The corrected AFK farm labels from this morning's forum review now show in the app.** The AFK sustain math no longer credits a heal on auto-fire while the damage math spends that same single slot on an attack: sustain is scored fully passive, the way AFK farming actually runs, and Momentum-gated attacks (Whirling Smash, Follow Through) no longer count toward AFK at all. The labels now read: Spines / Fiery Aura +0x8 passive (re-certified with better power choices), Titan Weapons / Bio Armor +3x8 passive, Fiery Aura / Fiery Melee +2x8 passive. These are floors for the combos, not ceilings — the open question of certifying to the AFK tier itself is tracked publicly in the known-gaps list. Thanks again to Maelwys.

0.12.42 — 2026-08-17 — The promised re-certification, and Adaptation done right

- **The full re-certification promised in 0.12.41 is complete.** Every one of the 28 recommended builds re-ran the full overnight search under the corrected model. Seven found

genuinely better power choices and are replaced — the Peacebringer Dwarf and Triform forms, the Warshade Dwarf, the Crab Spider Soldier, Sentinel Fire/Willpower, Blaster Fire/Energy, and Stalker Radiation/Dark Armor — and all seven ship with their exact certified slotting. The other 21 defended their titles against a fresh challenger and stand unchanged.

- **Set-unlock powers are now taken for you.** Bio Armor's Adaptation, Staff Fighting's Staff Mastery, Dual Pistols' Swap Ammo and friends are real picks that switch on your set's core mechanic and cost no enhancement slots — but the optimizer never chose them and the leveling guide never offered them. Now: generated builds take them automatically, the leveling guide offers them at their level, builds that lost their stances in the 0.12.41 cleanup receive the real Adaptation in a freed seat, and builds simply missing one get a note in the Build Assistant pointing at it.

0.12.41 — 2026-08-16 — Honest AFK farming, and set mechanics done right

- **AFK farm builds are now judged on damage nobody has to be there for.** The optimizer's AFK math was always strict about survival, but it still credited full attack chains no AFK farmer can cast. Now AFK certification counts damage auras, exactly one power on auto-fire, and nothing else. All three AFK farm builds were re-certified from scratch under the corrected model, and their labels state the honest tier: Spines / Fiery Aura keeps +4x8; Titan Weapons / Bio Armor certifies at +1x8 with +4x8 stated as unreachable; Fiery Aura / Fiery Melee lands at +3x8. Thanks to Maelwys on the Homecoming forums for the report that drove this.
- **A full re-certification of every recommended build is running after this release.** The corrected model changes nothing measurable for the non-farm builds (verified: zero score movement), but the full search is re-running for all of them as belt and braces; any build that improves ships in the next update.
- **Powers your set grants for free are no longer treated as picks.** Bio Armor's three Adaptation stances (and the same class everywhere: Staff Fighting's Forms, Dual Pistols' ammo types, Seismic Shockwaves) were being offered and even auto-picked as if they cost a power choice. The game grants these automatically with their power set; the planner now never spends a pick on them, and flags older builds that did.
- **Picks that accept no enhancements now work the way the game has them.** Bio Armor's Adaptation, Dual Pistols' Swap Ammo, Staff Fighting's Staff Mastery and friends are real picks with zero enhancement slots. They were hidden from the catalogue entirely; now they can be taken, show no slot row, and refuse enhancements — exactly as in game.
- **The "changes between builds" list can no longer show a previous character.** A slower computer could still see the last character's numbers leak into a new character's first report (a request started on the old character finishing after the switch). Responses from a character you have already left are now discarded, and a freshly generated character's first solve shows no change-list at all — there is nothing it changed from.

0.12.40 — 2026-08-15 — Three field-report fixes

- **A new character truly starts clean.** Starting a new character right after working on another could show a long "what changed" list comparing the new build against the

previous character's numbers. The previous character's measurements are now cleared the moment you switch, so the first solve of a new character reports only that character.

- **Saving right before closing is respected.** Pressing Save and then closing the window could still ask "Save before closing?" even though nothing was unsaved. The close prompt now re-checks your actual saved state and only asks when there is really something to lose.
- **Level 1 picks are seated the way the game seats them.** Some builds showed two level-1 powers from the same power set. In the game, character creation gives you one pick from your primary set and one from your secondary — the planner now always seats level 1 that way, and self-heals builds that were saved with the old seating.

0.12.39 — 2026-08-15 — Certified builds, served as certified

- **Four new fire-farm alternative champions.** The farm roster grows from one build to five, each a genuinely different way to farm: Tanker Fiery Aura / Fiery Melee (AFK), Brute Titan Weapons / Bio Organic Armor (AFK), Scrapper Radiation Melee / Radiation Armor (active), and Stalker Claws / Energy Aura (active) — none overlapping the existing Brute Spines / Fiery Aura, and all certified on current game data.
- **Champions now arrive with their certified slotting.** Until now the tool stored each champion's power picks and re-derived the slotting fresh on every visit, so the build you received could differ from the build the certification actually measured. Eighteen champions now carry their exact certified slotting and hand it to you unchanged — what the numbers describe is what you get. The moment you change anything (your own targets, your own earned slots, a different role), the solver takes over and builds for your ask, exactly as before.
- **A long-standing double-count in the game data is fixed.** 845 powers carried duplicate effect rows (an artifact of how conditional variants like critical hits and Containment parse), and the model summed the copies — pool and epic attacks were the most inflated (Boxing read roughly three times its real damage). 1,970 duplicate rows are removed; every displayed number and every score now sits on the corrected data.
- **Every champion re-certified on the corrected data.** The full power-pick search re-ran for all 27 existing contexts; ten champions found genuinely better power choices and were superseded (both Arachnos builds, Battle Axe Brute, two Kheldian forms, Invulnerability Tanker, Water/Kinetics Corruptor, and three farm builds). The other seventeen incumbents beat their challengers and stand unchanged.
- **One-slot powers explain themselves.** Cards holding a single set piece or a single proc now say what that slot is doing and why, like every other slotting decision.

0.12.38 — 2026-08-11 — The window that says why

- **A window that cannot start now says why.** If the engine fails to launch (most often an antivirus quarantining part of a fresh update), the window used to sit dark and empty. It now shows what went wrong, names the log file that holds the full reason, and says what to do about it. Reported by Glacier Peak on the forum.

0.12.37 — 2026-08-10 — Help, shortcuts, and a faster front door

- **Launching shows a loading screen in about a second.** Double-clicking used to mean nearly four seconds of nothing while the game database loaded before any window could exist. A branded loading screen now appears almost immediately, hands over to the app window, and the app takes over the moment the engine is ready. No fake progress bar; an honest spinner and one line saying what is happening.
- **The app opens on Powers & Slots, every time.** It used to reopen on the Leveling Guide for leveling characters. The road still introduces itself once, when a new 1-50 character is created; launching lands on the first tab.
- **Boomerang Slice is back, and it was never gone from the game.** The 0.12.36 notes retracted it as not live; that was wrong for this one power. It has been live since the February 10 update (Issue 28 Page 3), as Maelwys pointed out on the forum with the patch note in hand — thank you. The restored record matches the game's stated numbers exactly: 1.83s cast, 8s recharge, 30ft reach in a 30 degree arc, a 15-second resistance debuff, and it remains mutually exclusive with Slice, which the tool enforces. The 15-second Rending Slice bonus is shown but deliberately not scored yet. Wind Control and the Gadgetry and Utility Belt pools appear in no patch notes and stay out.

- **Search help, inside the app.** Help menu Search help: start typing (solve, lock, exemplar, set bonuses, import…) and the most likely term appears first with its one-line answer, other candidates beneath — arrow keys choose, Enter opens the full page. That page gives four plain answers — what it does, how it works, why it exists, and where it sits in the workflow of building, updating, tweaking or manually changing a build — plus a concrete example and, where it helps, a picture taken straight from the guided tour's example build. Every page ends with "go work on it directly": one click leaves help and lands on the real surface or the exact tour step. Opening help with nothing typed shows the four-step loop itself. 45 topics at launch; the PDF user guide stays for long-form reading.
- **Keyboard shortcuts for every major menu action.** Ctrl+S saves, Ctrl+I imports, Ctrl+E exports to Mids, Ctrl+K opens Search help, Ctrl+Z undoes (asking first, naming what it takes back), and Ctrl+1 through Ctrl+4 switch tabs. The combos are shown on the menu items themselves, ignored while you are typing, and documented in help — search "shortcuts".

0.12.36 — 2026-08-10 — The second accuracy pass

Six model versions of accuracy work (v39 to v44) and a full re-certification of all 24 bundled champions. Everything below is engine and data work against the game's own client files — and every claim in this entry was re-verified independently before release, after the retraction described at the bottom.

Things the tool was not counting, and now is

- **Critical hits.** Scrapper and Stalker criticals had never been scored in any version. Counted at the chance the game states per power, taking the lower of the two rates (the one against minions).
- **Domination.** Control powers last 50 percent longer while it is up, priced

at the uptime the inherent's own numbers imply.

- **Self damage buffs.** Build Up, Aim, Rage, Follow Up, Soul Drain and more.
- **Absorb shields**, **defence debuff resistance** (178 powers, which Super Reflexes had been denied entirely), and **slow resistance from powers**.
- **A dead lookup fixed:** the engine asked for an enhancement aspect called Recharge, which no enhancement carries. Recharge slotting was reaching none of your timed pets or click uptimes.

Corrections

- Three Soul Drain records had the wrong target cap or radius.
- Twenty-five control powers had the wrong duration or magnitude.
- Thirteen pairs of mutually exclusive powers are now refused, as the game does.

Your builds

All 24 champions re-certified across three waves. Every one is legal in game and records which model version certified it.

The Spines/Fiery Aura farm champion scores lower than before. That is a more accurate price, not a worse build.

Stated and deliberately not counted

- **Mez protection** is in the data and on screen for 229 powers but does not affect any score. It needs a decision about how often content mezzes you.
- **Criticals** use the lower of the two stated chances; the higher needs a mix of enemy ranks the model does not track.
- **Domination's magnitude bonus** is not counted, only its duration.

RETRACTED BEFORE RELEASE

An earlier draft of this release claimed Wind Control, the Gadgetry and Utility Belt power pools, and Boomerang Slice. **None of them are live in the game.** They were added from client files that carry unreleased content, all 34 records have been removed, and no version shipped with them. A new standing check now compares the tool's data against the last shipped release, so content the game does not offer can never enter again unnoticed.

Game data is unchanged (Homecoming 2026.1.1242).

0.12.35 — 2026-08-07 — Every set is clickable

Game data is unchanged (Homecoming 2026.1.1242) and the optimizer's model stays at v38, so no build's score moves and nothing was re-certified. Both fixes come from one field report.

- **Fixed: enhancement sets with an apostrophe in the name could not be slotted by hand.** Clicking Gaussian's Synchronized Fire-Control, Basilisk's Gaze, Achilles' Heel, Cupid's Crush or

any of the archetype sets in the picker did nothing at all, and left a syntax error in the console. The apostrophe was ending the click handler early. 40 set and piece names are affected, and this had been true since the first release. The solver could always slot them, so it only bit you if you picked one yourself.

- **The improvement report pointed at an export button that was not there.** It said to use "Export to Mids Reborn above" when that item lives in the Character menu. It now names the menu, and carries its own export button so you do not have to go looking.

0.12.34 — 2026-08-07 — Your picks stay yours

Game data is unchanged (Homecoming 2026.1.1242) and the optimizer's model stays at v38, so no build's score moves and nothing was re-certified. This release exists for one bug, reported from the field the same evening 0.12.33 went out.

- **Fixed: changing your epic pool could leave your build short of power picks, and nothing said so.** Switching epic pools and choosing "Switch and refill" is supposed to take the new pool's powers into the seats the old pool left behind. It could come back having filled none of them, and the build was then slotted anyway — so all 67 enhancement slots went into the powers that were left, the tally read 67 of 67, and the missing picks were invisible unless you counted them. Three things changed: the refill now draws its suggestions from the power pools your build actually holds (it was proposing powers from pools you had not taken and then discarding them, which is how the seats stayed empty); it checks afterwards that the seats really are filled; and if it still cannot fill them it says so and leaves your slotting alone instead of quietly solving an incomplete build.
- **The level-by-level guide now shows power picks you have not made.** A build short of the game's 24 picks used to render nothing at all at those levels, so the one page whose job is to say what happens at each level was silent about the picks still waiting for you. Open seats are now listed where they belong.

0.12.33 — 2026-08-07 — Knowing what to do next

Game data is unchanged (Homecoming 2026.1.1242) and the optimizer's model stays at v38, so no build's score moves. One power record was corrected — Tactical Arrow's Gymnastics, below — and no certified build uses it, so nothing was re-certified. Everything else here is what the app tells you and how it behaves.

- **The Build Assistant and the Stats page now say what they are for.** Both tools explained themselves in the smallest, dimmest text on the page, and the Assistant's explanation was being folded behind a "more" link and cut off mid-word, so the one paragraph describing the feature was the one paragraph you could not read. Both now open with a proper introduction at a readable size: what the tool does to a build you already have, what it will never change, and what you get back. Stats says that every number is a button, that you can price a single enhancement by pulling it out, compare every legal replacement before committing, and undo anything.

- **Gymnastics is back in Tactical Arrow.** Blaster Tactical Arrow was listing "Oil Slick Arrow" twice and never listing Gymnastics. The defence passive had picked up the wrong name and the wrong costs — a 90 second recharge and 15.6 endurance, where the real power is 10

seconds and 0.13 — and could not hold a defence set. It now reads correctly. No other power was affected and no build's score changes.

- **The app now says what order to work in.** Once a character exists, Powers & Slots opens with the four steps: say what you want more of, press Solve, tune one piece at a time on Stats, and change powers last. Each step takes you to the surface that does it. Before this the loop was written down in four separate places at four different depths, and the Solve button — the one control that redoes your build — sat nearly four screens below where you land. The guided tour teaches the same order, and what each step hands back.

- **Changing your epic pool can finish the job for you.** Swapping the epic used to remove the old pool's powers and stop, leaving you to find the new pool's powers in the catalogue, seat them, and run Solve yourself — while changing a primary or secondary set had offered to rebuild for you all along. The confirm now leads with **Switch and refill**: it takes the new pool's powers into the seats that just came free and re-slots toward your goal, in one action. The rest of your build is kept either way, and **Switch, I'll pick them** is still there if you want to choose them yourself.

- **Opening a character no longer carries the last one's state with it.** A sweep after the fix below found more of the same: the Stats breakdown stayed open on the previous character's power (click an attack on one character, open another, and the panel still stood there headed with a power the new character does not have), and the import comparison, the last confirmed solve, the generated build results and the incarnate recommendations all survived the swap as well. Every one of them now resets with the character. Your converter haul is deliberately left alone — it is a list you typed, not something the app worked out, and losing it on a character swap would be worse than keeping it.

- **Opening a character no longer counts as an edit.** A "What changed" panel was appearing on its own at launch, reporting a change nobody had made. Underneath it was something more serious: simply opening a character was filling the undo history, so the Undo button sat lit before you had touched anything — and pressing it could empty the build you had just opened. Loading a character now records nothing, and each character starts with a clean undo history so Undo can never reach back into the one you just closed. Real edits are unaffected: they still record, still report, and still undo.

- **Three panels on Powers & Slots were missing their outline.** The Build Assistant, In-game commands and How set bonuses stack were the only blocks on that tab without the blue line around them, so they read as loose text rather than as panels. They match everything else now.

- **A few small arrows were showing as a broken box.** The "" character used in eleven places had no glyph in the app's font. They are plain arrows now.

- **The big action buttons are a sensible size again.** The Solve, Generate and Reset buttons in the Build Assistant were stretching to the full width of the window, which looked odd on a wide screen and made them harder to aim at. They now share one comfortable width, whatever your window size. On a narrow window nothing changes: they still fill the space they are in.

0.12.32 — 2026-08-06 — The Stats page becomes a workbench

Game data is unchanged (Homecoming 2026.1.1242) and the optimizer's model stays at v38, so no build's score moves. Everything here is what the app lets you see and do.

- **The enhancement picker will not let you break the game's rules.** It already stopped you putting the same set piece twice in one power. Now a unique enhancement you already have somewhere else in the build is greyed out too, and it tells you which power is holding it, so the rule is visible rather than discovered afterwards as an error. Luck of the Gambler's global recharge is deliberately still offered — the game does allow that one repeatedly.

- **The swap list tells you what each replacement would do.** Open the picker to change an enhancement and every option on the list shows what it would do to the stat you are working on and to your set bonuses — a gain or a shortfall, worked out by rebuilding your character with that piece in the slot rather than estimated. Pick a stat on the Stats tab first and the list compares against it; set-bonus changes show either way, because losing a set tier is the cost that is easiest to miss.

- **Change something on the Stats tab and it tells you what you just did.** A "What changed" panel appears with the real before and after on every number that moved, and an **Undo this change** button right there if you do not like it. It covers any change you make, not just the ones made from the enhancement panel.

- **Click any enhancement and see exactly what it is worth, right where you clicked.** On the Stats tab, hovering an enhancement in the miniature wall names it; clicking one opens a small panel beside it listing every number that piece is currently buying you, set bonuses included. Click the next enhancement and the panel follows, so you can compare several in a row and see which one you can most afford to give up. The stat breakdown you had open stays open underneath. **And you can act on it there**: the panel offers "Swap this enhancement" and "Remove it", so deciding and changing happen in one place. It is an ordinary edit, so Ctrl+Z takes it back. It is worked out by rebuilding your character without that one enhancement and comparing, so it is what you would actually lose by removing or replacing it, not an estimate. That matters more than it sounds: pulling a single defence piece can drop a whole set tier and cost you max hit points, and only a real comparison catches that.

- **The two green marks in the stat breakdown now explain themselves.** Clicking a stat rings the enhancements feeding it, and boxes the NAME of any power that grants it by itself with no enhancement involved. Those are two different things and only one of them was ever explained, in a footnote you would see only if you had already worked it out. There is a legend at the top of the breakdown now, showing both marks.

- **Every exploration badge in the game is now listed on the page, and its name is the button.** The Zones & Badges catalogue used to be 57 closed drawers: to get one badge's location you opened a zone and read down a list. All 390 badges are on the surface under their zone now, and clicking a badge's name copies the game's own `/thumbtack` command for it, the command the client registers for placing a marker on your minimap. Paste it into the chat box while you are in that zone. Each zone still opens for written directions to every badge in it and what each one commemorates.

- **The 8 badges we have no coordinates for are not pretend buttons.** They are listed as plain text and say why, rather than looking clickable and copying nothing.

- **25 badges were hiding their location.** Their coordinates were only drawn alongside the written directions, so any badge without a write-up showed no location at all, even though we had its exact position. Every badge with coordinates now shows them.

- **The Leveling Guide's Flashback view opens on Praetoria, so you see it.** Praetoria's zones run level 1 to 20, so previewing Flashback from a level-50 character left the art panel with nothing to show. It now lands on the first stop that has a Praetorian zone, and puts you back where you were standing when you switch away again — the preview never moves your place on your own road.

- **And when there is genuinely nothing to show, it says so honestly.** The art panel used to read "zone art pending" for a level with no zone at all, which read as a missing picture. Those are two different things now: art we do not have yet still says pending, and a level outside the content says that instead, naming the range it covers.

0.12.31 — 2026-08-05 — The desktop app

Game data is unchanged (Homecoming 2026.1.1242), the model stays at v38, and the bundled champions are the 24 certified in 0.12.30. Everything in this release is the app itself: it stops being a browser page and becomes a real desktop application, rebuilt around your review notes and field reports.

- **Hero Companion is a real desktop window now.** It opens in its own window, and closing that window closes the app — no icon down by your clock, no tray to hunt through. Nothing extra installs: it uses the web view that already ships with Windows 10 and 11. Every file the app produces goes through a real Save As dialog, your settings and theme survive every launch, and the About dialog names the exact build you are running.

- **The update check runs when the app starts.** It compares version numbers against the releases page and tells you only when there is something newer. Nothing about you or your builds is sent, and you can turn it off in Settings. "Start Hero Companion when I sign in to Windows" lives there too.

- **A portable copy is never silently converted to an installed one.** If you run the app from the unzipped folder, the update button now explains that updating means unzipping the new folder over the old one, and points at the download page — instead of quietly running the installer, which is what it used to do (thank you, BasiliskXVIII).

- **The app is organized into four tabs**: Powers & Slots, Stats, Leveling Guide, and Logging, under a build tile that names your character (there is a Name field now). Menus explain themselves: anything greyed out says why. The app zooms to fit your window instead of rearranging itself, no panel has its own scrollbar, and on smaller displays the two-column regions fold to one.

- **Powers & Slots works like the in-game respec**: picks in level order along the game's own ladder, offering only what the game offers at each level. What the rules forbid is greyed with the reason on it, never hidden. Prerequisite counts and pool rules come from the game's files. The Build Assistant and the epic/incarnate plan run full width under the builder, so its tables read as tables — the side column opens only when you ask for an enhancement's details.

- **You are asked, once, whether to share your play data with the Pulse Boards.** The question sits in the opening menu, where it cannot get between you and your character. It spells out exactly what is captured, what leaves your machine, and what the public board shows. Sharing stays off unless you say yes, either answer is remembered, and both are reversible from the Logging tab — where the Play Log's on state now offers "Turn it off" just as plainly as the off state offers "Turn it on".

- **Stats explains every number.** Every row is a name, a one-line plain-English meaning, and the number. Your archetype's inherent appears at the top as an honest stat: each mechanic marked counted, shown only, or not modeled, so you know what is in your numbers. The level warning carries its own level input, and Ctrl+Z asks before it acts and names the edit it will take back.

- **Exemplared view.** Set the level you will exemplar to and the whole app shows that truth: powers you would lose are badged, dead set bonuses zero out, surviving ones stay, and a banner totals the cost — including what fully attuned enhancements would save you. An opt-in solve can even build FOR that level. The View menu explains exemplaring in plain English first.

- **Four alignments, properly.** Hero, Vigilante, Rogue, and Villain each carry their own theme and their own wordmark, drawn as crisp text from the game-styled art sheet. Archetype emblems — the game's own art — mark your build and your saved characters. Every dialog is the app's own, in your alignment's colors; nothing says "127.0.0.1 says" anymore.

- **Roles grew up.** The role list groups what is natural for your archetype versus off-role (allowed, but it will fight you), every role states the real floors the optimizer holds for it, and the old "Mixed / Generalist" is now Split role: name the jobs you actually want, as many as you want, with shares that always total 100. A "What this build delivers today" panel puts real numbers on each named job, and the Enhancement Unslotter is right there when you want slots back.

- **The improvement report answers "what did that buy me?"** Every solve diffs before and after across every measured axis — typed defense and resistance, recharge, recovery, single-target and area damage, pet damage, every buff and debuff row — and now per power too: "Empty Clips +18 DPS", with pets credited to the power that summons them. When nothing moved, it says why.

- **Buff and debuff numbers read your slotting.** The panel used to price every effect unenhanced; now each effect is enhanced by its power's own enhancement in that aspect, both directions — a debuffer who slots for the role finally sees the role move. Recharge buffs and debuffs are credited too, settled from the game's own per-power enhancement lists. The few effects the game ships no enhancement for are named instead of guessed.

- **One import door, and it teaches.** The menu's two import entries became one panel that explains both routes: the in-game /build_save_file for a character you play, and the .mbd file for a build planned in Mids Reborn. The Mids round trip is verified piece-for-piece both ways, and special origins (Hamidon, Hydra, Titan, D-Sync) come back with their real names instead of internal codes. Swapping a powerset now rebuilds the build cleanly instead of leaving ghost picks behind.

- **The Leveling Guide is called that everywhere**, its side-content preview is truly per-visit (leaving the tab resets it, and choosing a real alignment always outranks a preview), and the Flashback tab explains itself.

- **The guided tour was rebuilt for all of it**: 63 steps over the real tabbed shell, still on a drawn example screen so it never touches your build, and its audit now checks that every step describes something that still exists.

- **94 Incarnate Alpha boosts got their icons** — the game's own art, pulled from a corner of the client files our extractor had been missing.

0.12.30 — 2026-07-31 — Accuracy pass

Game data is unchanged from 0.12.26 (Homecoming 2026.1.1242). The optimizer's model moves to v38 (pet hit chance), and every bundled champion build was re-certified against the corrected engine before this release.

- **Power prerequisites now come from the game's own rule, not our guess.** The tool used to work out how many other powers a pool or epic power needs from where it sat in the list. The game states it per power, and the two disagreed in both directions: travel powers like Fly, Teleport, Super Speed and Mystic Flight were being held back when the game lets you take them freely, while Weave, Group Fly, Wall of Force, Misdirection and a few others were allowed with one prerequisite when the game wants two. Every pool and epic power in the app now carries the game's own count — all 467 of them, checked against the rule the game actually runs, with a standing check that refuses to let a build run if any of them ever drifts.

- **The build checker now catches missing POOL prerequisites too.** It checked epic and ancillary pools but silently skipped the regular power pools, so taking Vengeance with only one other Leadership power drew no warning even though the game will not allow it. Now every pool is checked, with the same plain-English message telling you how many more powers you need and from where.

- **Mace Mastery no longer appears for Scrappers.** The game gives it to Stalkers; we were offering a set that does not exist for that archetype.

- **Better slotting when several layouts tie.** Once a build reaches the defense and resistance caps, many different slot arrangements score exactly the same, and the optimizer used to return whichever it happened to reach first — builds that looked equal on paper could differ by a fifth in real performance. It now breaks those ties by what you told it the build is for:

endurance sustain first, then damage for damage roles or protection above the cap for tanks. When it is not certain, it solves the build both ways and keeps whichever the full physics model rates higher, so the build you get is never the worse of the two.

- **Every bundled champion build re-certified.** All 24 came through the scoring gate again on the corrected engine; 8 came back genuinely better and were replaced, the other 16 kept their existing builds because the originals still scored higher. The biggest gains were on the form-based Kheldians and the Arachnos soldiers.

- **Exported .mbd files open in Mids Reborn again.** Mids reads the alignment field as a case-sensitive value ("Hero"), and our exports wrote the app's lowercase "hero" — so a build exported from Hero Companion crashed Mids Reborn on open with "Requested value 'hero' was not found" (thank you to the field reporter with the SS/Shield Brute). Exports now write the exact casing Mids expects, for every alignment. If you have an older exported file that won't open, changing "hero" to "Hero" inside it fixes it — or just re-export after this update.

- **The optimizer can now propose Hamidon Origins.** The classic endgame move — two or three enhancement aspects in one slot — was something you could always slot by hand, but the optimizer never offered. For endgame content (incarnate trials and farms) it now weighs HOs against set pieces where they genuinely win: three Ribosomes in a resistance toggle carry a full set's worth of resistance enhancement in half the slots, at the price of the set bonuses. Every HO the tool places carries its attain note ("from Hamidon raids or merit conversion"), the enhancement picker's guided-tour gained a card explaining the trade, and the help guide gained a Hamidon Origins section. Other content presets are untouched — a general or team solve is byte-identical to before.

- **Pet damage stops pretending pets always hit (model v38).** Pets and henchmen are now scored with their real chance to hit, at their own level: the game summons tier-1 henchmen two levels below you (tier 2 one, tier 3 at level), so against +4 enemies they miss a lot — and now the numbers say so. What buys it back is counted too: ToHit buffs that actually reach pets (Supremacy itself carries one, Tactics projects; a self-only power like Focused Accuracy does not) and accuracy slotted in the summon power. Every fact behind this came from the game client or, where the client is structurally silent, the wiki with a citation (docs/pet-tohit-sources.md). Pet rows in the offense panel now show each pet's level and accuracy, the ToHit buffs get their own named list, and the help guide gained a "How pet damage is counted" section. Mastermind and pet-heavy champion scores will re-certify in this batch's combined wave.

- **Every proc choice now explains itself.** When the optimizer puts damage procs in a power instead of a full set — or seats a resistance or Force Feedback proc — the power's panel now says why, in one sentence with both numbers: what the procs add every use, and what the replaced pieces would have added instead. The numbers come from the same engine that prices your build; nothing is re-estimated. A ? on the note jumps to a new guided-tour card ("Why these enhancements") that explains the trade and how to overrule it: slot the set back and lock the power.

0.12.29 — 2026-07-27 — The guided tour

Game data is unchanged from 0.12.26 (Homecoming 2026.1.1242) and the optimizer's model stays at v36, so every certified champion carries the same score. This release is the guided tour, a security hardening, and a better user guide.

- **A guided tour of the whole app.** New here, or never sure what a button does? A compass in the header opens a chooser: pick one section or take the complete tour. The tour walks a drawn example screen — a Brute damage build with every panel filled in — so any part of it works at any time, even before you have loaded anything, and nothing about your own work is ever touched. Every card highlights the control it explains, you set the pace, and small controls get annotated drawings — which button is the padlock, where the cap line sits on a stat bar, which circular arrow checks for updates and which one leaves. A fresh install offers the tour once; after that it waits behind the compass and the ? beside each section — and every power card carries its own ?, which jumps straight to the card explained, controls labeled, no chapter to walk first. Every tour card also has "Save my spot & exit"; the compass offers to resume exactly there.

- **The user guide grew.** The Hero/Villain section now explains that switching sides also switches which accolades are assumed and recalculates your totals (it was described as pure style, which was wrong), and new "Locking a power" and "Reading the Numbers" sections cover the padlock and how to read the bars.

- **Security: another website can no longer make the app do things.** 0.12.28 stopped other sites *reading* from the app. This closes the other half: a page in another tab could still *send* the app an instruction your browser would faithfully deliver, such as telling it to close, even though that page never saw a reply. The app now refuses anything that changes something unless the request came from its own page. Nothing about normal use changes, and the tools and Companion Lite that talk to it are unaffected.

0.12.28 — 2026-07-27 — The security release

Five security fixes and one field-reported bug. Nothing about how you use the app changes, and nothing here alters build results: game data is unchanged from 0.12.26 (Homecoming 2026.1.1242) and the optimizer's model stays at v36, so every certified champion carries the same score.

Automated code scanning now runs on this project continuously. The security fixes below came out of its first pass, and none of them were being exploited: they were found by looking, not by anyone getting hurt.

- **Security: names from outside your machine can no longer inject anything into the app.** Text the app did not write itself, an imported build's name, an in-game export, and above all other players' character names picked up by the Play Log, was not being fully escaped before being placed into the page. A name containing a double quote could break out and attach itself to the interface. Now escaped properly. Names still display exactly as they are written, quotes and all.

- **Security: a search you type can no longer run as code.** The "nothing matches" message in the enhancement picker put your search text straight into the page without

escaping it first.

- **Security: the in-game build reader can no longer be walked out of your game folder.** It already refused paths outside your accounts folder, but the check compared the path as written rather than where it actually leads, so a shortcut-style link placed inside that folder could point anywhere on your drive and still pass. It now resolves links before deciding.

- **Security: the app no longer sends internal error details to the page.** When something failed, the raw error, including folder paths from your machine, was handed back to whatever asked. You now get a plain explanation of what went wrong and what to do, and the technical detail stays on your machine.

- **Security: other websites can no longer read your local Hero Companion data.** The app's own server was sending a permission header that let *any* website you happened to have open in your browser call it and read the answer while Hero Companion was running. That exposed your game install paths and account folder names, your saved characters, and your Play Log data, and let a page shut the app down. The app never needed that permission in the first place, since its page and its data come from the same place, so it has been removed. Nothing about how you use the app changes.

- **Backing out of the "new level 50" wizard takes you back to the start screen.** Picking an entry card and then changing your mind used to close the wizard onto an empty builder with no character in it, which looks like the app broke. Closing now returns you to "How do you want to start?", the screen you came in from. If you already had a character open, closing still returns you to it as before.

0.12.27 — 2026-07-26 — Targets you set, and honest advice when they fall short

Both fixes in this release came from reports in the forum thread. Game data is unchanged from 0.12.26 (Homecoming 2026.1.1242), and the optimizer's model stays at v36, so every certified champion carries the same score it did last release.

- **"Reset to imported build" no longer throws away your targets without saying so.** That button restores the build exactly as it came in, which also clears custom build targets and your front/back line and travel answers. It did that silently, so a solve afterwards quietly went back to the preset targets for your Content and Role, which reads as the tool overriding you. It now tells you what is about to be cleared and lets you back out, and if you go ahead it says what was cleared and where to set it again. Nothing prompts if you had nothing set.

- **When a target is out of reach, the tool now tells you what to do about it.** If a defense or resistance goal comes up short, you get the numbers *and* a concrete next move: the actual unpicked powers on your character that supply that stat and roughly what each would add ("adding Stealth (~7.0% base, more enhanced) and re-solving would close part of the gap"). When nothing is left that supplies it, it says so plainly instead of going quiet. Previously this advice only appeared if you had used the "Customize build targets" editor, so anyone who stated the same goal

through the wizard got a bare refusal with no suggestions. Two more faults fixed behind it: the Concealment pool was looked up under a name the game does not use, so Stealth could never be suggested, and a character with no unpicked candidates returned nothing at all rather than the honest "this ask may be unreachable on this pairing". This advice is deliberately limited to goals *you* set. The preset defense and resistance numbers behind a Content and Role choice are internal aiming points rather than promises, so the tool no longer offers to close a gap you never asked about.

0.12.26 — 2026-07-24 — Full power icons and in-app bug reports

- **Pulse Board updates reach phones on their own.** The board is a live page, but browsers (iOS Safari especially) were serving a cached copy after an update. It now sends a no-cache hint and reloads itself if the browser restores it from the back/forward cache, so the latest board shows up within a moment instead of needing a manual refresh.

- **Pulse Board section titles no longer clip on phones.** On a narrow screen the banner graphic (e.g. "INCARNATE TRIALS") was centred and scaled so the left side, where the title sits, ran off-screen ("...RIALS"). The banner art is now anchored to the left, so the title always shows and only the right-side flourish crops.

- **Play Log no longer shows the same account twice.** If you have two Homecoming installs, the same account folder can exist under both in different casing (`FiloFinFain` / `filofinfain`), which made two accounts appear as four "watch" chips. Account logins are case-insensitive, so these are now collapsed to one chip each — keeping the install you're actually playing (most recently active).

- **Report a bug without a GitHub account.** The button now opens a short in-app form — what happened, an optional email if you want a reply — and sends it straight to the developer. Your app version, model, game data and character are attached automatically, with an option to include your full build. Nothing leaves your machine until you press Send. (If the reporting service is ever unreachable, it falls back to the old GitHub issue link.)

- **Incarnate powers show their real icon on the tray.** Ageless, Ion, Control and the other slotted incarnates were drawing a plain glyph in the tray layout (the summary row already showed the real art); the tray now uses the same icon.
- **Filled in the missing power icons.** Newer powersets (Bio Armor, Psionic Armor, Staff Fighting, Dual Pistols ammo, Marine Affinity, Seismic Blast, Stone Armor and many more) were showing blank. Most icons were already bundled, just unreferenced; the rest were pulled straight from the game client's own texture files. Power-icon coverage went up by over a thousand entries; the tray/builder gap for real build powers dropped from 469 to 38 (the remainder are inherents and a few hidden/auto powers the game itself doesn't give an icon).

- **Removed the "Power colors & glow .powerCust" export.** A leftover from the shelved costume/appearance idea, never verified in-game. Gone from the import screen.

- **The Leveling Journey no longer covers a new level-50 build.** The road auto-opens only for a character still leveling to 50 — a fresh 1-50 plan, or an imported character below

50. A brand-new level-50 kit goes straight to the build page. Reopen the road any time with the Journey button.

- **Clearer Leveling Journey header.** The auto-open checkbox now sits on one line with its full label right beside it ("Open this road automatically while leveling to 50") — it was previously clipped, then briefly wrapped into a narrow column. The alignment switcher now reads "Preview another side: [buttons] (your character is unchanged)" so the reassurance makes sense. The step-by-step link reads "Switch to the step-by-step list view", and the "Want to switch sides?" alignment help shows a "click to expand" cue.
- **The Leveling Journey now shows the character you're actually on.** Opening the road after switching to a different character used to show the previous one's powers — the road is now rebuilt for the current character every time it opens.
- **Clearer "start over" button.** The button at the top right now reads "Load another character, or make a new one".

0.12.25 — 2026-07-23 — The Journey grows up

The Leveling Journey went from a road with placeholder labels to a full 1–50 field guide. Everything below is on the road now.

- **Every zone shows its game picture.** The comic-panel splash plate for each zone (City Hall for Atlas Park, the Rogue Isles for Mercy Island) sits on the level's card, with the zone name printed across it. Pulled straight from the game client, 38 zones covered.
- **Where every badge is, in plain English.** Each exploration badge now carries its location and a written direction ("on top of the casino, 120 yards due east of Meteor Do"), plus its coordinates. Over 350 badges, with a pointer to the VidiotMaps in-game overlay for the visual finder. Badge location data by Gulbasaur's community effort via the public-domain content database.
- **When each Task Force and Strike Force opens.** Every TF, SF, and trial on the road shows "available at N" — Positron at 8, Synapse at 15, Lord Recluse at 45, all fifty of them.
- **Master and Incarnate-trial challenge badges.** Each task force and iTrial shows its Master badge and the exact challenge sub-badges you need ("Master of the B.A.F. — earn: Not On My Watch, Alarm Raiser, Gotta Keep 'Em Separated, Strong & Pretty").
- **A readiness check for every part of a zone.** Each neighborhood is read against your level using the game's own con system: "Challenger Technology 21–27, tough" if you can go now, "come back at 27" if it is over your head. So you know not just where to go, but whether you are ready for it.
- **Every zone that fits your level, at a glance.** At any level the road lists all the zones open to you, computed from the full 47-zone level table — not just the ones on a written route.
- **Where to level, level by level.** Each stop names the zones and story arcs worth doing there, in order, with the XP-pause tips that keep you from outlevelling a story — a hero and a villain route, condensed from the community "Good Missions" and "Mean Missions" guides by Gulbasaur.

- **See any side's content.** A Hero / Vigilante / Rogue / Villain / Flashback switch on the road lets you browse the other side's leveling path without touching your character — it is preview only, your build never changes. A "Want to switch sides?" note explains how Null the Gull actually does it in-game, and the Flashback view explains Ouroboros replay and its level requirements.
- **Every zone's foes, level range, and a wiki link.** Each zone entry names who you will fight there and links to its full wiki page.
- **Bigger, clearer, and it lights up.** Larger fonts throughout, the zone title in the game's own heavy comic-title style, and the level you are looking at glows gold on the road.

Earlier in this cycle:

- **The Journey opens the moment your 1–50 build finishes**, for every new or resumed leveling character. Closing it is just closing it; the header switch ("open automatically for new characters") turns the auto-open off, remembered and reversible.
- **The road drags like a map** — grab and pan; card clicks still work.
- **Open cards never hide under the title bar**, and the panel scrolls in its own window instead of stretching the road.
- **Accolades on the Journey** — a drawer of the build-affecting accolades with their permanent bonuses and how to earn them.

0.12.24 — 2026-07-22

- **The Leveling Journey.** Your whole 1–50 as a road you scroll like a map — every level a stop with its power pick, new enhancement slots, and milestones; click a card for what that level buys you (stat gains and tips). Security-level badges sit at 10/20/30/40/50 with the game's own descriptions, and a "Zones & badges" drawer holds the full exploration-badge catalog straight from the game's files, find-hints included. The **Journey button in the header is its on/off switch** — click to open, click again to close (and Esc work too, and the button glows while the road is out). Zone display names, level ranges, and task force suggestions are labeled as coming (they need server-side game data, in progress) — nothing on the road is a guess.
- **A 1–50 character's Journey starts ON.** Build or resume a leveling character and the road opens itself the first time, introducing what it's for, how to close and reopen it, and how to turn on the game's chat log — with the full 2-minute setup steps right there — so your Play Log fills while you play. Then it's your call: close it once (or press "Got it") and it never opens itself again. Builds that aren't leveling see the same introduction the first time they open the road themselves.
- **The wizard never asks a question twice.** Starting a new character used to show the "tell me your aim" recommender and the "How do you play?" section with the same three questions — anyone picking their own archetype saw both. The recommender now owns no questions of its own: it reads your Role / Mostly-in / You-fight-from answers and lives right below them ("Not sure what to roll? … Find my character"). Asking it with Role unanswered flags the question instead of quietly assuming damage.
- **Farm advice works for the real farm choices again.** When the generic "Fire Farm" option was split into AFK and Active in 0.12.22, eight pieces of server-side farm guidance

kept listening for the old name only — the archetype recommender's farm ranking, farm build warnings, flame-epic preference, farm incarnate advice, and the proc treatment for support-role farm builds had all quietly stopped applying. All of them now recognize both farm choices (and the old name, for older saved characters).

- **Your archetype's hidden mechanics are now in the math AND on the screen (model v36).** Fury, Domination, Defiance, Vigilance, Scourge, Containment and the other built-in meters were silently absent from scoring; each is now credited with its stated steady-state assumption, and a new **Inherent Mechanics** panel shows every credited meter with the assumption written next to it — you see the mechanic, not just absorb it. Champions were re-certified under the new math (one build improved enough to be replaced).
- **Live game-log connections are now a named list, not a bare number.** The connections display previously could count stale or rotated log files as "live" (the four-connections-for-two-accounts field report). Each connection now shows its file path and last-activity time, and a log that stops being active stops being counted, on a stated timeout.
- **Optional: start Hero Companion automatically when you sign in to Windows.** Asked once on first run of the installed app — never turned on silently — and changeable any time from the tray menu ("Start automatically at login"). A login start stays quiet in the tray: no window or browser tab opens until you click. Uninstalling removes the setting completely. Same opt-in mechanism Companion Lite shipped in 0.1.18.

0.12.23 — 2026-07-21

- **Endurance is now real physics (model v35).** The old scoring gave an endurance-broken build 45–72% credit no matter how fast the blue bar emptied. Now the model plays the fight out: full output until the bar drains, then only what recovery can pay for — measured against each fight's actual length. A build that empties in 17 seconds keeps most of its score in short scraps and collapses to its true ~11% output in a 4-minute Archvillain fight, exactly as it plays. Sustainable builds are never penalized, and a build the tool generates can no longer "afford" a rotation it cannot actually run. (This keeps the promise made in the forum thread: endurance was named as the known gap, and this is the fix.)
- **No hidden incarnate assumptions.** Scores and warnings now use your BARE recovery — the old model quietly assumed incarnate-era recovery (Ageless) in some content. If your build includes incarnates via the toggle, the ledger says so in plain text instead of assuming silently.
- **Endurance procs are worth their real, measured value.** Performance Shifter and Panacea's +endurance procs were priced at conservative guesses; they're now credited at their measured average from real game logs (about 0.53 and 0.40 endurance/second — the measurement matched the game's proc formula to within 1%). Theft of Essence is priced from the formula with its usage assumption stated. (Correction found during measurement: Power Transfer's proc is a self-heal, not an endurance return — it stays priced as healing.)
- **+Max Endurance finally counts.** 40 set bonuses granting +Max Endurance existed in the data but were silently dropped; they now raise your endurance pool in totals and in the fight math, alongside accolades.

- **Travel toggles are never silently ignored again.** If you run Fly or Hover, the ledger always shows that drain. Declare a ranged/hover playstyle and the fight math counts it too (a grounded melee build still doesn't pay for a travel power it turns off).
- **Cardiac gets recommended when it should be.** If your build's endurance ledger runs a deficit, the incarnate recommender now leads with Cardiac and explains the trade — more damage on paper loses to damage you actually have the endurance to deliver.
- **Your typed-in build targets can't be traded away.** A declared custom target (like "45% fire defense") now outranks every internal heuristic in the optimizer — the endurance repricing exposed a case where the solver traded 0.4% of a declared defense ask for recovery value. Declared asks are promises; a pinned test now holds the line.
- **The Build Assistant knows whether you're creating or retooling.** With an imported or resumed build on screen the panel now reads "Improve this build", explains that it re-slots what you have, and shows the build's existing identity instead of re-asking setup questions. Every control says what it does in plain words — the steer chips say they're nudges, not commands, and the buttons name their consequence ("Re-slot everything except your 3 locked powers").
- **"Preserve my IO sets" is now visible padlocks on the power cards.** Turn it on and every power you hand-slotted shows a — a locked power's slotting is never altered by any solve, byte for byte (tested every release). Unlock individual powers to let a re-solve improve them. The checkbox and the padlocks can never disagree: it's checked when everything is locked, unchecked when nothing is, and shows a dash for a mix.
- **Hand edits update totals instantly.** Changing an individual IO always recalculated immediately — now the totals panel visibly pulses so you can see it happen. There is no apply button for hand edits, by design.
- **One targets control instead of two look-alikes.** Once you set custom targets, the chip itself shows what you asked for ("Your targets: Fire def 45 · Fire res 90 — edit"), clicking it reopens the editor, and removing them asks first and says what happens (back to the preset).
- **Custom targets now shape the power PICKS, not just the slotting.** Ask an Invulnerability Tanker for 45% defense and the proposal now includes Tough Hide (the defense auto the old picker treated as filler) — and tells you what it gave up to serve the ask. Anything not fully reachable is reported with numbers and a concrete suggestion ("adding Weave would close part of the gap"), never silently traded. The targets editor also states what clearing a field means: you drop the demand, not the stat.

0.12.22 — 2026-07-20

- **This is the first digitally signed Hero Companion release.** The installer and app now carry a Microsoft-verified signature ("Joel Andrew Chambers"), so Windows shows a named publisher instead of an "unknown publisher" warning. Every release from here on ships signed. (SmartScreen's "not commonly downloaded" prompt is reputation-based and fades as downloads accrue — that part takes time, not more paperwork.)
- **The engine now values a Mastermind's pet buffs (model v34).** A Mastermind's main damage is its pets, and the buffs it casts on them — Supremacy (+25%), Accelerate

Metabolism-class ally buffs, Temporal Selection (single-target, on your biggest pet), Beast Mastery's Pack Mentality (assuming 8 of 10 stacks with pets engaged, stated on the build) — now raise pet damage in every score and label. Caster-only buffs (like the Musculature Alpha) correctly do not. A new "Pet damage buffs" section on the offense panel shows exactly which buffs applied, at what strength and uptime. Stated simplification: pets are modeled as always hitting; pet accuracy is a named next step. Two champions genuinely improved under the corrected math and were re-certified (the Radiation/Sonic and Radiation/Radiation Defenders); every other certified build re-verified with its score updated. Making the search fully exploit the new term is the named next engineering item.

- **Every power, powerset, and epic pool now uses the game's current names.** The client's own files are the authority — 74 power names and 4 pool names that had drifted (the old "Field Mastery" is the game's "Energy Mastery", "Breath of Fire" is "Fiery Breath", and so on) now match what you see in game. A standing check keeps them matched. The suspected "phantom epic pool" turned out to be a real Mastermind ancillary wearing its stale name — every epic offered to every archetype was verified against the game's own eligibility rules (zero leaks; a standing check now pins that too).
- **No more silent dead pages.** If the app's local server isn't running (or a page action fails), the page now says so plainly with a Reload button instead of buttons that silently do nothing — the field report that found this also added an interaction test to every release so it can't ship again. Uncaught page errors surface the same visible banner, and older saved characters that a newer version can't fully read load safely with an honest note instead of deadening the page.
- **Imported builds with empty slots say so.** An import now flags "in progress" powers ("4 empty slots: Dark Consumption") so an unfinished character never reads as a finished plan.
- **Honest slotting labels.** "Full set" now means the set is actually complete. A partial set mixed with procs reads "Frankenslot"; a clean partial set (plus universal globals like Luck of the Gambler) reads "Partial set". The globals list also learned knockback protection (Karma, Blessing of the Zephyr), slow resistance (Winter's Gift), and the Theft of Essence +endurance proc.
- **The content picker's farm section is now two honest choices** — AFK Fire Farm (passive) and Active Fire Farm. The old generic "Fire Farm" is retired; saved builds that used it get a prompt to pick the one that matches how they play (never silently remapped).
- **Four more dead-air fixes from release-night testing.** The build button now names the exact field you're missing and highlights it in red instead of repeating gray text; the Solve button does the same when Content isn't picked, and honestly says "no changes" when your slotting already meets the goal. The biggest one: Solve's "are you sure about this role?" question was being asked in a hidden panel on every installed copy — Solve would silently wait forever for an answer you couldn't see. It now asks where you can see it. And the import file dialog no longer strands you on an empty page if you cancel it. A set's own proc piece now counts toward its "Full set" label (a 6-piece Javelin Volley is a full set, not a mix).
- Data currency: game-client data re-exported 2026-07-15; current through the game's July 7 patch.

0.12.21 — 2026-07-17

- **The AFK fire-farm champion's label was corrected upward — publicly owed, now delivered.** The game's June 23 patch made Temperature Protection's +MaxHP and +Regeneration enhanceable; our data snapshot predated it. With the data re-synced to the current client, the same shipped Spines/Fiery Aura build certifies at **+4x8, sustaining 42.5 HP/s** (the old label said +3x8). Thanks to Maelwys for the correction. A new structural check now diffs effect existence and enhanceability against the client so this class of staleness can't hide again.
- **Accolades are no longer invisible.** Below the powers and IO sets you'll find every build-affecting accolade from the game data, with a search box. Checking one moves your build's real numbers; a press-and-hold preview shows what all of them together would do without committing anything. Each row's shows the game's own description, and 26 of 28 explain how to earn it (full badge chains on 20); the rest say plainly that the requirements aren't documented from game data yet. New level-50 builds assume the four standard accolades and say so right on the build — untick any you don't have and the totals follow. Nothing is ever applied without you choosing it. And it follows the game's own rule: every accolade is hero-side, villain-side, or neither, read straight from the game data — and your character is one alignment, so the other side's accolades are greyed out as unavailable, exactly as in game (Portal Jockey and Born In Battle are the same bonus, one hero, one villain). Switch sides with the alignment button and the panel flips and re-assumes that side's standard set. The accolades that aren't side-locked stack normally, and so does every distinct bonus.
- **The numbers now say what's in them.** A line above Build Vitals states exactly which assumptions are folded in (accolades applied, incarnates on or off), and named contribution lines appear beneath the stats they touch (" Accolades +321 HP", " Alpha (Musculature) +45.0% damage") — read from the engine's own ledgers, never re-guessed on the page.
- **Incarnate choices our math doesn't price yet are marked "not yet modeled"** in the picker instead of silently doing nothing.
- **The Unrelenting Fury regen proc is priced** in the AFK sustain ledger (it was invisible to the sustain math before), and the active-farm objective now treats survival as hard requirements rather than a score — damage throughput decides the picks, exactly as active farmers build.
- **Certificates state exactly what they prove.** Every certified build's card now says its score comes from a converged search and prints the canonical number, without claiming more than that.
- **Fixes from Joel's field walks:** power-card summary strips render at full height again (no more crushed green sliver); a failed accolade load now says "Couldn't load — try again" instead of silently disabling the feature forever; accolade checkmarks no longer leak between characters; and every build entry point (new character, optimize, respec) applies the standard-accolade preselect consistently.
- **Every power card now shows what reaches its own numbers.** Open a power's and, where a global buff actually multiplies into that power's damage (an Alpha like Musculature, a Hybrid, both together), a named line states the amount folded in — and where the game's damage cap holds it below the raw buff, the card says the effective value, not the raw one. Every card also carries an

honest footer: build-wide bonuses such as accolades don't change a single power's numbers, so they're pointed to Build Vitals rather than faked onto the card. Untick the incarnate preview and the line and the numbers drop together. All of it is read from the engine's own ledger — when several incarnates stack, each is named, never lumped under one.

- **Incarnates and Epics are treated by their real unlock rules.** While you're leveling a character, the tool now warns when you preview endgame content you haven't earned yet: incarnate abilities unlock at level 50, and Epic / Ancillary powers at level 35 (Patron pools also need their Patron arc completed). Nothing is blocked — you can still preview your finished build — but a plain note tells you these aren't available at your current level, and the totals say "endgame preview" instead of implying you have them. A level-50 build never sees the warning.
- **Iron Man's accolade tells you how to earn it, sourced from the game.** It's the one-million-damage badge (shown as Adamant, Iron Man, or Ironwoman depending on your alignment and body type); the pop-up now states that, confirmed against the game's own text.

Companion Lite 0.1.18 — 2026-07-17 — The signed install

- **A real Windows installer.** Companion Lite now ships as a proper per-user install (Start Menu entry, clean uninstall, no admin prompt) instead of a loose exe. Under the hood it moved from a single packed file to a folder layout, which starts faster and stops tripping antivirus "packer" heuristics.
- **Signed with a verified publisher.** The installer and app are code-signed, so Windows shows "Joel Andrew Chambers" as the publisher instead of an "unknown publisher" warning. (SmartScreen's "not commonly downloaded" notice fades on its own as more people install — signing removes the scary warning immediately, and reputation does the rest.)
- **Auto-start is your choice.** On first run it asks, once, whether to start with Windows — never silently on — and you can flip it any time from the tray. Uninstalling removes it cleanly.

Companion Lite 0.1.17 — 2026-07-15

- **The feed tells you the truth when it can't upload.** Errors now lead with the human reason ("getaddrinfo failed" means no network/DNS, "timed out", "connection refused") instead of a bare class name, and the tray status shows a visible retry line — "RETRYING, N failures since HH:MM" — so a dead feed is never two green icons over silence.
- **One machine, one uploader.** When the full Hero Companion app runs alongside Lite, exactly one of them feeds the board (a shared lock — the same pattern that already keeps them from double-capturing). No more racing the upload bookmark.
- **Your remembered "no" now covers Lite too.** Turning the feed off in the full app's Play Log silences Lite's uploads as well — one choice, honored everywhere, reversible in the same place. The tray says plainly when the feed is off and why.

0.12.20 — 2026-07-15

- **Custom build targets now survive to the finish line.** When you set your own numbers (the fire-farm 45% defense / 90% resistance case), no later optimization step may trade them away — the solver meets your ask and the finishing passes are held to it. (Work order A: the proc pass was breaking a target-serving set after the solve had already met the number.)
- **Repeated Optimize presses are stable and honest.** The same button now gives the same build from the first press, and when your imported character owns fewer slots than the level-50 plan places, the result says so plainly instead of silently showing slots you don't have yet. (The reported "slot inflation" was measured everywhere and never existed — slot conservation is now a standing audit anyway.)
- **No impossible defense numbers.** Where a total includes out-of-combat stealth defense (Hide's big AoE layer), the fight value prints right beside it — "90% 30% in combat" — with the explanation on hover.
- **High-cap archetypes get their real resistance ceiling.** Tanker and Brute resistance targets were quietly clamped to 75% in some solve paths (their real cap is 90%); Kheldians and Arachnos soldiers were clamped under their 85%. Every target now uses your archetype's actual cap.
- **The public Pulse board is honest about BOTH kinds of quiet.** The page now shows "Data through <time>" always, warns "no new game data since <time>" when the feed goes quiet, and separately warns when the render pipeline itself stops. (Field lesson: a healthy renderer over a dead feed kept re-stamping the page "fresh" while the numbers froze for 19 hours.)

- **Peacebringers and Warshades: pick your form, get its champion.** The level-50 wizard asks Kheldians one more question — Human, Dwarf, Nova, or all three (tri-form) — and serves that form's own certified champion build as the base to build under. Each form champion is a committed build for the way you actually play, honestly labeled: form-swapping play isn't modeled yet, so each certificate covers exactly what it names. Other archetypes never see the question.
- **The Kheldian champions are here — and the whole roster passed a harder legality bar to ship.** Eight new certified builds (Peacebringer and Warshade, all four forms each), plus re-converged Night Widow and Water/Kinetics Corruptor. Every champion now has to be BUILDABLE from level 1 exactly as picked — the game grants your third power pick at level 2, and builds with nothing legal to take there are refused. That gate caught and rebuilt several champions, including two that shipped in 0.12.19. Twenty-four certified reference builds ship in this release.
- **A certified AFK fire-farm champion (Spines/Fiery Aura Brute) — with an honest difficulty label.** Two new Content choices in the wizard: AFK Fire Farm (passive — the build must hold 45% Fire defense, capped Fire resistance, and out-heal the spawn while you're away from the keyboard) and Active Fire Farm (you're at the wheel; the build spends its budget on damage). The AFK champion's certificate states exactly what it sustains: certified at +3x8 — the +4x8 asteroid worst case is out of reach for this combo, and the label says so with the numbers instead of hoping you won't notice. (An interruptible heal can never count toward AFK sustain — the game cancels the cast when you're hit, so the math refuses it too.)
- **Damage procs in auras and patch powers now count.** Procs slotted in a damage aura (Blazing Aura, Quills) or a ground patch's summoning power (Irradiated Ground) were

priced at zero since launch; they now earn their keep on the pulse rate the game client's own records state. The active fire-farm champion is deliberately NOT in this release: our own field measurements say these procs may be worth even more than the formula gives, and that champion waits until the numbers are settled rather than shipping a build we'd have to walk back.

- **Reset means reset — every start-over path.** The upper-right and every "start over" entry point now genuinely clear every answer, preset, and leftover; the wizard opens blank. (Found in the field: one path kept old answers alive.)
- **Nobody ever needs Ctrl+F5 again.** Every release and dev restart reaches your browser on a plain reload — fixes can't hide behind a stale cache.
- **Power cards read like power cards.** The full power name owns the top line (no more "Radioac…"), and the details button is always visible instead of appearing only when you hover over where it would be.
- **Clicking Build never looks like nothing happened.** A large "Your build is ready" button leads the result and scrolls into view — and while a build is still baking, a live seconds counter now ticks instead of silence. Most combinations solve in about a second; some are genuinely hard math, and after a few seconds the counter says plainly that it's still working, not stuck. No fake progress bars — the old "(~1s)" promise is gone because it wasn't always true.
- **The header finally shows a version you care about.** It leads with the app version; click it for a plain-language About — what the app is, what a "build model" version means, where the game data comes from, how many certified reference builds you're running with, and the links that matter (forum thread, releases, Pulse Boards, credits).

0.12.19 — 2026-07-13

- **Eight new champion builds — nearly every archetype now has a certified anchor.** Brute (Battle Axe/Fiery Aura), Tanker (Invulnerability/Super Strength), Scrapper (Broad Sword/Super Reflexes), Stalker (Radiation Melee/Dark Armor), Sentinel (Fire Blast/Willpower), Blaster (Fire Blast/Energy Manipulation), Dominator (Mind Control/Fiery Assault), and Arachnos Soldier (Crab Spider) join the five existing champions — thirteen converged, certified level-50 builds served automatically when you pick those combinations. Peacebringer, Warshade, Night Widow, and one Corruptor combo arrive later this week (the Kheldians with their full form set).

- **Everything Companion Lite does, the full app now does too.** The Play Log tab gains the Pulse Boards: "My private board" builds your personal board (scorecards, market ledger, raids seen) from your own capture store, right in the app — local data, never uploaded. "What sharing shows" previews the sanitized public variant, so the choice to share is an informed one. And the live-board feed itself: an explicit, reversible opt-in behind the full terms (shown in the app before anything ever uploads), with the same privacy guarantees as Lite — account names replaced with meaningless codes before upload, private messages never captured, no machine details ever. Capture itself was always shared between the two apps; if you run Lite alongside, they coordinate exactly as before.

- **"Mixed role / Generalist" now appears everywhere a role is asked.** It was

always in the main Role dropdowns, but the "Start a new character" discovery flow ("I want to…") never offered it — now "do a bit of everything (generalist)" is a first-class answer there, with its own archetype recommendations (the sets-span-roles picks, Kheldians included). Also fixed on the way: choosing "control / lock down" or "command pets" in discovery used to silently blank the wizard's Role question while marking it answered — control now carries over as Controller/Lockdown, and pets honestly leaves Role for you to pick (commanding pets is an archetype, not a role).

0.12.18 — 2026-07-10

- **Your core armor toggles are enhanced like they deserve (model v30).** The optimizer used to treat a met survival target as "done" — past that point, enhancing your strongest shields was worth literally nothing to it, so powers like Fire Shield or Temp Invulnerability could end up as two-slot parking spots for global IOs while a free minor power got a full set. That threshold is gone: survival keeps its real, continuously-measured value all the way to your archetype's caps, so the strongest armor powers get their resistance and defense aspects enhanced first — with endurance reduction riding along — and the weaker or free powers take the mule duty. Global IOs now also prefer the cheapest hosts, so an expensive toggle like Weave keeps its slots for real enhancement. (Thanks to Maelwys for the precise field report that pinned this down.)
- **Ten set-bonus families the planner literally could not see now count (model v30):** knockback protection, slow resistance, all six mez-duration bonuses (confuse, hold, stun, sleep, immobilize, fear), movement speed, range, and endurance discount — 103 bonus tiers back-filled straight from the game client's own data, verified value-for-value. They appear in your totals and on every enhancement card (the "not yet in totals" note is gone), and the optimizer's scoring now values knockback protection, slow resistance, and your controls' longer mez durations. The three stackable knockback IOs (Karma, Steadfast Protection, Blessing of the Zephyr) are priced too — mag 4 each, exactly as the game grants them.
- **Preview toggles now stack — every combination the game allows.** The one-preview-at-a-time rule blocked real play: Build Up + Aim is the classic opener, Farsight + Group Invisibility + Power Boost is how a Time build checks its soft cap, Meltdown + Shadow Meld is a genuine defensive layer. All of it now stacks freely. Honesty replaced restriction: short strike windows (Build Up class) wear their own chip, and while any are checked the totals panel says loudly that you're looking at a BURST VIEW — a window of seconds, not what the build sustains. Cycling buffs (Hasten, Farsight class) keep the chip with their real uptime at your build's recharge — and a short-window power your recharge makes effectively permanent (Parry spam) earns the cycle label its math deserves. Also fixed: unchecking behavior could leave a chip LOOKING checked while its power no longer counted — chips now always repaint with their true state.

0.12.17 — 2026-07-10

- **The "count in totals" checkbox now matches what the game actually allows.** It used to be one uniform Σ checkbox on every power — which lied for most of them. Now there are three honest states, each with its own color and glyph (never color alone): always-on powers (autos, passives, inherents) show a locked badge with no control at all, since the game gives them no

off-switch; toggles keep a checkbox (, green, default ON) as their real home — uncheck a mule host or run a quick "what if Weave were off"; timed self-buff clicks (Hasten, Build Up, and their kind) get a preview toggle (, orange, default OFF) that previews one window at a time, since the game only ever has one of these running as a meaningful "on" state — checking one switches any other preview off. Plain attacks get no chip at all. Also: the endurance-sustainability warning now respects your checkbox choices (an unchecked toggle no longer counts against your endurance), and it actually shows up in the totals panel when your checked toggles + attack chain would run you dry.

- **Customize the targets the solver chases.** "Customize build targets…" (on the play-style summary and next to the Content/Role pickers) opens an editor seeded from your chosen preset: typed AND positional defense, per-type resistance, recharge, recovery, regeneration, and max HP. Set a number to chase it, clear it to drop it — a resistance-primary set like Fiery Aura can finally chase 90% S/L/F resistance plus recharge and HP instead of being held to defense-shaped preset thresholds. Values clamp to game reality (your archetype's own resistance cap); when the ask exceeds what 67 slots can buy, the solve gets as close as it can and says so. Custom targets persist with the save, and you can keep them as named presets of your own to reuse on any build. Honest labeling throughout: a custom-target build is YOURS — it never reads as a certified champion build. Also fixed on the way: resuming a saved character now restores its Content/Role picks into the dropdowns (they used to come back empty, so a re-solve could silently run against different targets).

- **The same set piece can no longer be slotted twice in one power.** The game doesn't allow it, and now neither does the picker: pieces already in the power show " slotted here" and can't be picked again. If a build arrives with duplicates anyway, validation flags it as a real error, the duplicate no longer conjures set-bonus tiers (tiers count distinct pieces, so the card's "N of 6" and your totals always agree), and stackable globals like Luck of the Gambler +recharge now respect the game's rule of five — a sixth copy grants nothing, exactly like in-game. Identical Hamidon/Titan/Hydra Origins and D-Syncs still stack freely (the game's actual rule).

- **Preview Enhancement Boosters before you buy them.** Every boostable enhancement's detail card now carries a booster stepper: walk it from +1 to +5 one level at a time and watch your real totals move — "+3 reaches the soft cap, +4 and +5 buy nothing here" is now something you can see instead of guess. Previewed pieces show their level exactly like imports do ("50+3") but in a distinct preview color, and the totals panel says plainly that previews are not saved as owned. Pieces that can't boost say why: attuned enhancements (including ATOs and Winter sets) scale with your level instead and keep exemplaring down, and Hamidon/Titan/Hydra Origins and D-Syncs get their strength from their own level. The card also explains where boosters come from and the exemplar trade-off in plain English.

- **Power Boost previews for real.** Powers that amplify your other buffs (Power Boost, Power Build Up) now have their own preview chip (): switch one on and your buffable defense and ToHit values amplify exactly the way the game does it — effects the game marks "Ignores Buffs" are skipped, verified per-effect against the game's own data. Combine it with one buff preview (Power Boost + Farsight) to answer the real question: does this build actually hit the soft cap in its burst window? The totals panel labels the result as a burst view, never sustained. (Clarion Radial isn't modeled

yet — its incarnate data doesn't carry a verifiable amplifier record; the tool says so rather than guessing.)

- **Running toggles are no longer called "mules."** A toggle or always-on power that hosts global uniques (Fire Shield with a Steadfast, Weave with a Luck of the Gambler) now reads "Global host — this toggle runs; its slots carry build-wide bonuses" instead of "Global mules," which implied the power was dead weight. True set-real-estate powers keep honest mule wording.

- **The detail card grew into a real rail.** The right-side panel now scrolls on its own (a long six-piece card browses in place while your build stays on screen), every slotted enhancement carries its own — hover the chit — so the full detail card is one click away without entering the change-picker, and both rail views are LIVE: slot a new piece of the displayed set and the roster chips and lit bonus tiers update in place, no reopening. The power info panel now shows the full merchant-style set view too: one set in a power renders its card inline; a mixed-set power gets one compact row per set ("Shield Breaker 1/6 — next: +2.5% recovery") that expands to the full card. Set-bonus tiers also read cleaner — repeated per-type values collapse to one line, and bonuses the game grants but the totals can't count yet (slow resist, movement, range, mez duration) say so honestly instead of showing a bare name.

- **The true in-game enhancement experience.** Click a slotted enhancement and open its full detail card: the game's own description text with honest numbers at YOUR piece's level and boost state ("Enhances the defense debuff potential of a power by 17.4%…"), the attuned and unique rules, and the complete parent set — every piece marked slotted-here / elsewhere / missing, and every set-bonus tier lit as you attain it, with " next piece" showing exactly what one more piece buys. All text extracted from the game client itself (227 sets, every piece and tier), not paraphrased. The picker now shows what's currently slotted with a "Full details" button, and power cards reveal a hint so the info panels are discoverable.

0.12.16 — 2026-07-09

- **Every "How do you play?" question explains itself the moment you answer it.** The Role and "You fight from" pop-ups were silent on the first pass through the wizard (they come before "Mostly in", and the explainers waited for that answer). Now each question pops its tailored explanation as soon as YOU answer it, on every surface that asks — Start New, Respec 50, and Change how you started — while the combined summary still waits until your Role and Content are actually chosen. Also: the import screen now says plainly that in-game /build_save_file exports carry no boost levels — import a Mids .mbd to bring +1..+5 and HO levels across.

- **Henchmen inherit your set bonuses — and the optimizer knows.** The game gives Mastermind henchmen half of your true set bonuses (verified straight from the game client): +Max HP arrives as a flat amount from the Mastermind's own hit points, identical for every henchman, so your squishy tier-1s gain the most staying power — exactly the effect the forum discussion described. Builds are now scored with each henchman's real survival in the fight (tier hit points from the game's own tables, deaths cost the actual resummon and re-upgrade cast times), so +HP, defense and resistance bonuses carry honest extra value on a Mastermind. Globals like Unbreakable Guard and accolade HP do

NOT reach henchmen (also the game's rule), and the planner never credits them there.

- **Heal-strength set bonuses exist now.** The game's 11 heal-strength bonuses (Numina's 4-piece +6% heal, Doctored Wounds +4%, Panacea's 6-piece +6%…) had parsed to nothing — same story as the accuracy bonuses fixed in 0.12.15. All 11 are restored from the game client, the totals panel shows your +% Heal strength, and the optimizer values what they do to your actual healing output.

- **Slot anything by hand.** The enhancement picker now offers the single enhancements a power accepts alongside its sets: common crafted IOs, Hamidon / Titan / Hydra Origins, and D-Syncs. Identical copies stack freely (that's the game's rule for these), totals price them exactly, and they round-trip through Mids export/import.
- **In-combat view (suppression).** A new Stats toggle shows your totals as they are mid-fight: powers like Stealth lose their suppressible defense the moment you attack or get hit, exactly per each effect's own suppression flags — same as Mids' Options > Effects and Maths > Suppression. Display only: builds are optimized on the same numbers either way.
- **Real enhancement levels from imports.** Boosted IOs (50+5) and level-53 Hamidon Origins now import with their true levels, show as "50+5" / "53" on the slot, and count in the totals at the game's real math (+5% per boost level; a 53 HO is worth 38.3%, not 33.3%). Fixed a one-off: Mids stores levels 0-based, so every imported IO used to read one level low. Exports to Mids now carry boosters back too.
- **Cards say where the res job went.** A base-slotted power that carries its own resistance debuff (Melt Armor and friends) now names the power actually holding the res procs and why it won the job, instead of a vague "budget went elsewhere."
- **No more duplicate epic pools.** Three pools leaked into the wrong archetypes' dropdowns by stale upstream data (Stalkers and Dominators each saw two same-named "Mastery" pools). The game client's own eligibility rules say who gets what; the extra entries are gone.

0.12.15 — 2026-07-08

- **Fresh gold-standard champions, certified on verified game data.** All five champion builds were re-converged from scratch under this release's corrected model and data — every value they stand on now checks against the game client itself. Old champion scores looked bigger; these are the honest numbers.
- **The wizard never chooses for you.** All four "How do you play?" questions start unanswered and the Build button waits until you've answered them — the planner never invents your playstyle. Each answer is tagged with where it came from ("your pick" / "from your setup"), and reopening the wizard restores what you chose — including your travel power. Travel is never auto-picked: endgame content entry can require specific travel (BAF and Lambda admit only Flight or Teleport), so that call is always yours.
- **New Role option: Mixed role / Generalist.** The honest choice when you don't specialize — balanced targets for your content, judged by overall contribution instead of one role's lens. Its pop-up explains the trade plainly.

- **Slotting notes are easier to find and use.** The whole note chip on a power card is now the button (hover for the bonus summary, click for the reasoning), with a one-time hint on your first build. Power-card tooltips lead with the full power name, so truncated names are readable. The / + slot buttons explain the shared 67-slot pool, and the slot counter tells you when you have free slots to spend. The power-info panel closes on Esc or a click outside.
- **Global Accuracy set bonuses now exist — and are valued.** A parser gap had dropped every global-accuracy set bonus in the game (Luck of the Gambler's 4-piece +9%, Adjusted Targeting, the archetype sets' +15%…) — 65 bonuses, invisible to the totals and the optimizer alike. All 65 are restored straight from the game client's own data, the totals panel shows your +% Accuracy, and the optimizer now prices accuracy by the game's to-hit math: against +3/+4 enemies (iTrials, hard content) accuracy multiplies your real damage until you reach the 95% hit ceiling, so builds for that content chase it — and builds for even-level content don't waste slots on it. The data reality-check now verifies 282 set-bonus values against the live game (a name-matching bug had silently limited it to 43), all matching.
- **The "How do you play?" questions now explain themselves.** Every choice in the build wizard pops a detailed explanation written for YOUR character — a Mastermind's "Damage dealer" talks about henchmen and pet sets, a Super Reflexes scrapper's iTrial preset explains why it chases positional instead of typed defense. A new summary panel shows exactly what your combined answers make the planner chase (the real defense, resistance, and recharge targets), so nothing about the build's direction is a mystery before you hit Build.
- **Credits: Maelwys.** His expert reviews on the Homecoming forums caught what our own tests missed, round after round. Added to the in-app credits and CREDITS.md.

- **Common IO icons are back.** 0.12.13 accidentally dropped every common IO's icon, so a generated build's Accuracy/Damage/Endurance IOs rendered as empty-looking slots next to expensive globals. The same defect gave Hamidon Origins a level they don't have, which scaled imported low-level HOs toward zero. Both fixed, and the audit now fails hard if any slotted piece ever ships icon-less again.
- **Hamidon Origins stack legally.** The validator warned "duplicate piece slotted 2x" on double-Nucleolus cores. HOs, Titan/Hydra Origins, and D-Syncs are not set pieces — the game lets you slot as many identical copies as you like, and the planner now agrees.
- **Real attacks reach 6 slots.** The optimizer always wanted six, but a budget accounting gap plus an order-blind trim kept every real attack at exactly five while filler powers kept six. Attacks now get their sixth slot; any trimming falls on junk fills and filler powers first, never the rotation.
- **Proc bombs carry accuracy.** Big proc bombs reserve a slot for a Nucleolus Exposure (Accuracy/Damage, recharge-free so every proc keeps its full chance) — and bombs no longer shrink a 6-slot power down to the proc count.
- **No more orphaned set pieces.** The res anchor, Force Feedback seating, and the endurance-relief pass all refuse swaps that would strand a lone set piece with no bonus, eat a global, or overwrite a proc. Force Feedback now lands in a spammed knockback attack's spare slot instead of cannibalizing a proc bomb.
- **Signature support buffs always work.** A damage-role Mastermind's Barrier Reef

shipped as a bare 1-slot mule; signature buff clicks from a support set now always carry a working set, whatever role the build solves for.

- **Max HP totals are honest.** A /Regen build displayed "+58913% Max HP" (flat hit points summed as a percent). Totals now show the real number — "+79.9% = 2410 HP (capped; 3286 uncapped)" — plus regeneration in HP/sec and recovery in end/sec, the way the game reports them.
- **+HP set bonuses were 10x too big.** Verified against the game client's own set-bonus powers: a "Large Increased Health Bonus" is ~1.88% of base HP, not 18.75%. Every +HP set bonus was inflated tenfold in the totals and in the optimizer's HP math. Two Touch of the Nictus pieces now read +1.87%, a third Luck of the Gambler +1.12% — matching the live game exactly.
- **Card notes now describe the actual decision.** Slotting chips say "5 of 6 Red Fortune — earns +5% recharge, +1.5% fire resistance" instead of a vague "full set for its bonuses"; a couple of procs is labeled "Procs", not "Proc bomb"; and Hamidon-Origin proc hybrids get their own "Proc hybrid" note explaining the pattern (these cards previously showed nothing and could wrongly trigger the respec nag — that misfire is gone).

0.12.14 — July 7, 2026

- **Synced with the game's July 7 update (Issue 28, Page 3).** The patch normalized Dark Armor's Obscure Sustenance to a 60 second recharge for every archetype; our data carried the old 180 seconds for Brute, Scrapper, Stalker, and Tanker. A full diff of all 10,708 powers against the patched client confirmed these four values were the patch's only data change — archetype tables, pet classes, and proc rates all verified unchanged.

0.12.13 — July 7, 2026

- **Resistance procs for every role that can carry them.** Achilles' Heel, Annihilation, and Fury of the Gladiator multiply the whole spawn's incoming damage — and a damage role owns the biggest share of that damage. The optimizer now hunts a home for each res proc in every build (debuff toggles first, then the biggest eligible power), not just on debuffers and controllers. Annihilation's proc rate was also corrected against the game's own data.

- **Force Feedback: Chance for +Recharge finally counts.** The famous knockback proc is seated in your spammiest knockback attack and honestly valued: the average recharge it sustains (chance per cast × 5 seconds ÷ how often you actually fire) flows into everything recharge touches — attack chains, Hasten uptime, debuff cycling, pet resummons. The build totals show exactly how much of your recharge it carries.

- **Hamidon Origin enhancements modeled.** All 62 special enhancements are now priced correctly when slotted, and the optimizer uses the master pattern: a proc-hybrid attack's filler core trades up to two Nucleolus Exposures (about 66% accuracy and 66% damage in two slots, with no recharge to depress the proc rates). Premium set cores are left intact — those bonuses are build-defining.

- **Champion builds re-converged** under the corrected model and shipping in this update, plus a fix to pet resummon timing that slightly undercredited global recharge on timed summons.

0.12.12 — July 7, 2026

- **Henchmen and pets priced like the live game (the big Mastermind fix).** Pet damage was built on a stale snapshot that rated henchmen more than twice as hard-hitting as the real game, counted a single pet where the game spawns a squad, and treated every summon as permanent. All three are fixed from the game's own data: henchman damage now uses the live tables, Soldiers count as 2 Soldiers + 1 Medic (tier squads of 3/2/1 across all six Mastermind primaries), timed summons like Spiderlings earn only the share of a fight they're actually up (recharge slotting shortens the resummon wait), and powers whose pets don't inherit slotting no longer get phantom enhancement. The optimizer values pets on the squad's real, uptime-weighted output — the "healing mule" Mastermind builds die here.

- **Champion builds re-converged under the corrected model.** Every gold-standard champion was re-solved to full convergence with the accurate pet math and ships in this update.

- **Live play capture improvements** (shared with Companion Lite): the server pulse counts formations rather than repeated recruiting shouts, Positron Parts 1 and 2 are distinguished in every spelling, your characters' servers are detected automatically from the game's roster file, and the recruitment-channel list ships in the updatable content pack.

- **Your server shows on the board — detected automatically.** The game itself keeps a roster file (playerslot.txt) naming each character's server; Lite looks your characters up there the moment they appear in the log, so the live board's "Servers reporting" line fills in with zero configuration. Only the characters actually seen in your logs are recorded — the roster itself is never stored or uploaded.

- **Positron Parts 1 and 2 finally count separately.** Recruiting-alias matching is now longest-first, so "posi 2" no longer collapses into generic Positron — in every spelling recruiters use (posi2, pos 1, 1st posi, ...). The same fix protects every multi-word alias (ice mistral, mort kal, kmitf).

- **The recruitment-channel list ships in the content pack**, so shard global channels (where iTrials and raids organize) can be added as data. Private channels (tells, whispers) remain excluded unconditionally.

Companion Lite 0.1.13 — July 7, 2026

- **Companion Lite now feeds the live Pulse Boards directly.** No accounts, no tokens, no setup, no publish button: while it runs, captured play uploads over HTTPS to the project's locked storage and the board pipeline imports it into the live page within minutes. Uploads are incremental (only new play), tagged by an anonymous install id — account login names are replaced with meaningless codes before anything leaves the machine, and machine names or paths are never read at

all. The only thing that ever becomes public is the rendered board itself.

- **The terms are the consent.** First start shows the terms (also under About): using Companion Lite means you accept that your captured play feeds the live board. If you don't accept, quit and uninstall. Turning game logging off (/logchat) stops capture any time.

- **A simpler tray.** Open the live Pulse Board, the in-game logging setup, Status, About (terms), Check for updates, Quit. The local board page and every publish or connect control are gone — Lite captures and feeds, the live board is the board.

Companion Lite 0.1.12 — July 7, 2026

- **Your Pulse Board now lives online, not on the game machine.** On the owner's machine (the one with a publish token), Companion Lite keeps the live web page current automatically — publishing at most every 15 minutes and only when new play was captured — and "Open Pulse Boards" opens the online page directly. Machines without a token are untouched: their board stays a private local page and Lite never talks to the network.

- **Published boards are scrubbed before they leave the machine.** The public variant drops the local file path and never shows account login names — an unnamed character appears as "Unnamed character #1" until capture learns its name. Character names (which are public in game) are kept.

- **The server pulse counts formations, not shouts.** A recruiter repeating "lf2m dfb" until the team fills is one formation, not a sighting per shout: repeated asks for the same content collapse into one, the recruiter saying "full" closes it (their next ask is a new run), and 10 minutes of silence does too. Dual-boxing is covered by the same rule — two of your clients logging the same shout count it once.

- **Money stays off the public board.** Influence earned/spent and market flow show only on your private local board — how much anyone makes is not community data. The future price board will use per-item prices, never personal wealth.

- **Levels, badges, and merits wait for the character sync.** A fresh capture only sees a sliver of a multi-year character, so achievement state stays off the public board until a one-time in-game character sync (level, badges, accolades, vet levels) can bring a character over whole. Public character cards show play since capture began: days seen, XP, drops, defeats.

- **"Recent formations" actually shows recent ones.** The card was frozen on the first 20 sightings ever captured; it now rolls with the latest 20.

0.12.11 — July 6, 2026

- **Sharper end-game builds from a fully re-converged optimizer.** The best-known build for each context was re-solved to true convergence under the current model and the corrected proc data, so the head start the tool builds from is stronger.

- **Live play capture (opt-in, private messages excluded).** With chat logging on, the app turns your sessions into insight — your own rewards and drops, plus what's forming on your shard from the public recruitment channels. It learns your server's shorthand as it reads. Private messages (tells and whispers) are never captured, and nothing leaves your machine.

- **Long-recharge attacks and holds now get the proc treatment (model v25).** A single-target power like Seismic Smash or Dominate keeps a 2-3 piece accuracy/damage core and fills its remaining slots with damage procs when the proc math favors it — the classic expert pattern. The kept core avoids recharge pieces on purpose (slotted recharge lowers proc rates). Pet summons are never raided; their slots stay the pet's.

- **Proc rates corrected from the game itself.** Every damage proc's rate now comes straight from the current game data; 11 were wrong, including the archetype set procs, which were undervalued by up to 43%.

- **The respec suggestion now notices when the optimizer has learned.** Resuming a build made under an older version offers "Built under an older optimizer — see what's improved?" even when nothing is structurally wrong with the slotting. Build the plan to see whether a respec is worth it; if nothing meaningful improves, it says so. Saying "no" is remembered for that character until the optimizer actually changes again.

- **IO levels show on resumed builds again.** Resuming a saved character could show set pieces without the level badge under their icons (while common IOs still showed 50). Saves are now healed as they load, so every existing save gets its levels and icons back without re-saving.

0.12.10 — July 6, 2026

- **Pet summons are slotted for the pet now.** A summon power — Mastermind henchmen, a controller or Arachnos pet, Gang War, Phantom Army — was sometimes being used as a parking spot for the player's own always-on healing IOs (Numina, Miracle, Panacea) instead of a set that actually helps the pet. Those healing IOs belong in Health, where they buff you, so that's where they go now. Every pet summon is reserved for a real pet set — its accuracy, damage, and the pet enhancement bonuses (and the Mastermind archetype IOs) — and is forced to take one rather than sitting on filler. Thanks again to Maelwys for the example. Deeper slotting quality (procs versus full sets by attack, minus-resistance procs, when recharge in a summon actually matters) is still being tuned.

0.12.9 — July 6, 2026

- **Game data brought current.** Our numbers came from a Mids database that turned out to be about six months old, so recent balance patches were missing. We now reality-check against the live game and fixed everything that had drifted: the Brute, Sentinel, Tanker, and Dominator defense/resist/damage modifiers (this is why Brute defense was reading low), a batch of power endurance/recharge/range values, and slotting rules — including Mastermind archetype IOs now being slottable in personal attacks. Set bonuses were checked and are current. Thanks to Maelwys on the Homecoming forums for catching the stale data.

- **The slotting tags now show on any build, not just after Solve.** Resume or import a build and each power shows why it's slotted the way it is (full set / proc bomb / global mules) right away — previously those tags only appeared right after solving, which made a loaded build look like the update hadn't taken.

- **Respec worksheet.** When you resume a build with slots not earning set bonuses, a small bar appears at the top of the build — "Ready for respec?" — that opens the full plan as a pop-up (so it's there when you want it, out of the way when you don't). The plan shows every change power by power — the old slotting struck through, the new below it — the stat gains, and a grocery list split two ways: what to craft or buy, and what to unslot and sell (with a note on each item's worth). Every line has a checkbox to track your progress, and the whole thing is saved to the character, so you can work it over several sessions of crafting (the bar shows "N/M done"). Apply it to your build when ready, Undo to revert, or mark it completed. It never appears on a build that's already well slotted.

- **Explanations everywhere.** The slotting tags (proc bomb, full set, frankenslot, global mules) and the respec card now have a **?** that opens a plain-language explanation of what the term means, why that approach is used, and your options — including why, for example, a frankenslot can beat a single six-piece set.

- **Updating no longer leaves a ghost tray icon, for real this time.** When the app updates itself, it now removes its own tray icon cleanly before the installer takes over, instead of being force-killed and leaving a dead icon behind that only clears when you mouse over it. (0.12.8 improved this for extra copies; this closes the main path — the app replacing itself.)

0.12.8 — July 6, 2026

- **Watch multiple accounts at once — per-character stat cards.** Dual-boxers can now tick more than one account, and the Play Log shows a card per character side by side (Rattle's kills and haul next to your farmer's), each with its own fit link. Accounts are toggle chips — click to add or remove one from what's watched.

- **The Play Log moved up into the main workspace.** Instead of a detached strip far below everything, it now docks in the wide build column — filling that space when no build is loaded, and sitting neatly under the build when one is. On narrow screens it still stacks full-width like the rest.

- **Cleaner haul advice.** Incarnate salvage and crafting materials no longer get a "KEEP" tag — there's no decision to make (you bank them and spend later), so the haul now saves its keep/sell verdicts for the drops that actually warrant one (enhancement recipes and salvage). Incarnate materials are still counted in your loot summary.

- **The Play Log now knows who you're playing — and links to their fit.** It reads the character name from the log ("playing Rattle") instead of only showing the account, tracks it across character switches, and attributes your stats to the right character. If that character has a saved build in Hero Companion, a one-click **"load their fit"** link appears — so switching from Rattle to Lime Juice brings up Lime Juice's build, not whoever you had open. (The game only names your character on a fresh

login, so if it says "not detected yet," log out to character select and back in.) Because players sometimes rename characters, the fit link is one you can set explicitly — "link the open fit to this character" — so it sticks through a rename and fixes any wrong name-guess. No fit for a character yet? It offers to import their build straight from the game.

- **The Play Log's account switch is now obvious.** Which account you're watching shows as a row of chips (the active one highlighted with), and switching is one click on another chip — no more hunting for a "change account" link. For dual-boxers this matters: each account is its own log, so you flip between your two characters' stats instantly, and a note says so.

- **Your farm haul now knows your build.** When the Play Log is watching a character that has a saved fit, any drop whose set your build actually uses gets starred and floated to the top of the haul as a **"keep — for your build,"** naming the power that wants it. A standard set you would normally vendor becomes a keep when it is in your plan. Drops that do not fit still get the usual keep-or-sell advice.

- **Every power now explains how it is slotted.** Each power in a generated or solved build wears a small tag saying *why* it is slotted the way it is — a full set for its bonuses, a proc bomb (more AoE damage than a set here), global mules (uniques that each work from one slot), or a deliberate frankenslot. Hover it for the full reasoning. High-end slotting can look like scatter until you know the plan; now the build tells you the plan.

- **Fixed some warnings that were plain wrong.** The off-role check used to tell a Brute or Mastermind its powers "do not extend to damage dealer," which is nonsense — those are damage dealers and are now recognized as such. The "no damage enhancement" warning no longer fires on a power that is carrying a real set or damage procs. And Physical Perfection (and the epic-pool tier rules around it) now require the correct number of prerequisite powers, verified against a large corpus of real builds.

- **Better proc choices.** When the optimizer proc-bombs a power, it now ranks the available procs by actual expected damage, so the strongest procs land instead of whichever the data happened to list first.

- **The tray icon behaves.** Updating no longer leaves a "ghost" tray icon that only clears when you mouse over it — the old copy now shuts down cleanly and removes its own icon before the new one takes over. The tray menu also gains **Check for updates…** (it tells you right there whether a new version is out) alongside Open and Quit. Hero Companion keeps running in the tray after you close the browser tab so reopening is instant; the tray menu is how you drive or quit it.

0.12.7 — July 5, 2026

- **The Play Log parser is now built on real logs.** The first cut guessed at line formats; this version is validated against tens of thousands of lines of actual Homecoming chat logs. It reads XP and influence, enemies defeated (a real kill count), every drop sorted into recipes / salvage / incarnate materials / crafting mats with keep-or-sell advice, and Consignment House sales and purchases

— while correctly ignoring the wall of combat, heal, and chat spam around them. Recipe drops map to their enhancement set for a proper verdict. (Veteran levels aren't written to the chat log by the game, so post-50 progress is shown as the incarnate materials you earn instead — that's what the log actually records.)

- **The Play Log reads live and walks you through setup.** Once you pick an account it now updates on its own as you play (a " live" marker shows it's watching), so you don't have to keep clicking refresh. A built-in guide spells out the one-time in-game setup — make a chat tab named "Companion", add all channels to it, and turn on `/logchat` — so the only thing you have to remember is turning the log on. If there's no chat log for the day, the app says so and tells you how to start it. The opening screen is now framed simply: turn your sessions into *your* stats, all private and local — with any future community sharing kept as a separate choice. A plain privacy note spells out the promise: the tool only ever handles game data (prices, drops, stats), never your real name, email, location, IP, or account login, and never anything about the other players in your chat log.

- **The Play Log now asks before touching anything.** It read game files without asking and without saying what happens to the data — against this app's own rules. Now the section opens with a plain statement: what it reads (the chat logs your game writes), what you get (insights), and the promise — everything stays on this computer, nothing is uploaded or shared, and any future optional sharing will be a separate, clearly labeled, anonymous opt-in. Until you click "Enable the Play Log", the app doesn't even look at the game's folders. "No thanks" is remembered and reversible.

- **Fixed: the Play Log could dead-end.** Clicking "Read new log entries" before choosing an account replaced the account choices with an error message that told you to pick an account — with nothing left to pick. Now the read button simply doesn't exist until you're watching an account (it appears the moment you pick one), errors always bring the choices back, and the account choices look like buttons (" Watch filofinfain") instead of plain text.

0.12.6 — July 5, 2026

- **The Play Log (first cut).** A new full-width section at the bottom of the app reads your in-game chat logs (type `/logchat` in game once to start writing them) and turns them into insight — no raw log ever shown. Pick which account to watch (multi-account players choose their main), hit "Read new log entries" after a session, and get: session totals (XP, influence in/out, merits, defeats), progress (level-ups, badges), and your recent haul with keep-vs-convert/sell verdicts on every drop. Reading is incremental — only new lines since last time. The line formats are provisional until validated against real logs, and the tool says so: it reports how much it parsed and shows any data-looking lines it didn't recognize, so early users help perfect it.

0.12.5 — July 5, 2026

- **Power trays now follow the community standard** (researched from player forums): Tray 1 is your active rotation — openers, then the single-target chain in slots 1-3 (the muscle-memory keys), then AoEs and cones, with the nuke pinned at the end. Tray 2 is mid-fight clicks — self-buffs, then

heals, then endurance recovery grouped together at the end. Tray 3 is set-and-forget — toggles in switch-on order (your armors first, then pools), utility, and Rest parked last. Tray 4 is travel and sprints, deliberately away from the combat keys. The same kind of power lands in the same place on every character, so your muscle memory transfers between alts — the way experienced players actually set up their bars.

- **The Incarnates brick.** Your six incarnate choices (the solver's recommendations, or your own picks) now live in a card of their own — same shape as the in-game power trays and docked right above them, with the real in-game incarnate icons (family art + rarity ring) and the full ability name on hover. Slots you haven't filled show as dashed placeholders.

- **Every IO icon now shows its level.** Set pieces the optimizer slots (and ones you pick by hand) carry the level to buy or craft them at — the set's maximum, so a Cloud Senses piece reads 30, an Achilles' Heel proc 20, premium sets 50 — right under the icon, the same way common IOs already did.

- **Procs are named properly.** Slots the proc pass fills used to read just "Annihilation: proc" — they now carry the real piece name ("Annihilation: Chance for Res Debuff"), so you know exactly what to buy.

- **Ctrl+Z undoes your last edit.** Mis-clicked a slot? One keystroke takes it back — same as the Undo button (power picks, slot changes, enhancement swaps, pool changes). Typing in a text box keeps its normal Ctrl+Z.

0.12.4 — July 4, 2026

- **Level 1 always satisfies the creation requirement: one primary power AND one secondary power**, each from that set's first two — the basics come before anything else, on every archetype (same treatment as the epic-pool prerequisite rule). A Poison/Sonic Defender's level 1 is Shriek/Scream + Alkaloid/Envenom; Envenom-and-Alkaloid-both-at-1 can't happen anymore, anywhere it's displayed.
- **Old saves fix themselves.** A character saved by an earlier version could carry an impossible level-1 arrangement that survived every update because it was stored with the build. The app now re-checks the pick order every time it recalculates and quietly corrects it — open the build and it's right, no re-solve needed.

0.12.3 — July 4, 2026

- **Only one copy of the app can run.** Launching Hero Companion while it's already running now just opens the running copy's page instead of silently starting a second (or third) hidden copy on another port — the cause of "the app shows no difference after an update" reports: the browser was talking to a leftover old copy the whole time. After a self-update, any leftover old copies are shut down so the new version is the one you see.

- **Fixed: a generated build's pick levels were dropped when loading an

AI-designed tier**, so the build grid fell back to naive level badges (Alkaloid AND Envenom both shown at level 1). The seated pick levels now survive the load.

0.12.2 — July 4, 2026

- **Fixed: slots the game could never grant.** An enhancement slot is earned at a specific level and can only be placed in a power you already have — so a power picked at level 49 can hold at most 4 slots (its free one plus the 3 earned at 50), and the 47 + 49 picks share just 6. The optimizer now seats heavily-slotted powers earlier in the pick order, re-solves with tighter budgets when even reordering can't fit (dense melee epics), and the validator calls out any hand-made arrangement the game would refuse ("picked at level 49 — it can hold at most 4 slots").

- **Smoother one-click updates.** The tab you click "Update now" in simply becomes the new version when the install finishes — the app no longer opens a duplicate tab while the old one lingers on the old version. (If that tab was closed, a fresh one still opens after a short wait.)

- **Level 1 is now framed as character creation.** In game your level-1 powers are one of the first two powers of your primary AND one of the first two of your secondary (verified against 2,255 master builds) — the plan now always seats a legal creation pair at level 1, the leveling walk says which choice was made and what the other option was, and the validator flags a build that skipped both of a set's first two powers.

0.12.1 — July 4, 2026

- **The decorative frame corners are gone.** The hero/villain identity now lives entirely in the color themes; the app uses the full window edge to edge.

- **Fixed: epic-pool pets taken without their prerequisites.** The game requires two other powers from an epic pool before its top-tier powers (Ice Elemental, Summon Spiderlings…) and one before the mid tier — the optimizer now takes the lower powers first, the search respects the ladder, and the validator counts precisely ("requires 2 other Ice Mastery powers — this build has 1").

- **Fixed: inherent powers shown as picks.** Health, Stamina, and other inherents are granted automatically by the game — the respec order no longer tells you to "take" them at a level (they get their own "automatic" footer line with slot counts), and their cards wear an "auto" badge instead of a level. Real picks' derived levels no longer shift to make room for them.

0.12.0 — July 4, 2026

The layout release — a complete redesign around one idea: a tight, horizontal, Sidekick-shaped workspace where every block fits snugly and every gap holds useful information.

- **The powers wall**: your build is a grid of uniform icon cards — power icon, name, level badge, enhancement icons with IO levels — flowing in pick order. No columns, no holes.
- **The build summary course**: beneath the wall, three equal bricks — Build Vitals

(a labeled DEF/RES table plus recharge/recovery/HP/DPS), active Set Bonuses with stack counts and rule-of-five warnings, and the Uniques Carried checklist.

- **One tight masthead**: identity and icon-only actions in a single 40px strip.
- **Power Info panel**: click any power's name and a detail column opens on the right — type, availability, endurance cost, cast time, live damage/DPA/cycled-DPS (proc contributions included), allowed enhancement categories, and what's slotted in it right now. Numbers update with every change.
- **Horizontal-first layout**: character setup (archetype, sets, pools, epic, incarnates) is a compact left rail with Stats and the Build Assistant stacked beneath it; the powers icon grid owns the entire wide side. The masthead is one tight strip — identity, vitals, icon-only actions — that stays pinned as you scroll.
- **No AI anywhere.** The panel is simply the Build Assistant — deterministic presets, goals, and instant Solve. (The AI seam still exists for those who bring their own key, but it's opt-in and invisible otherwise.)
- **The build is now an icon grid** (Sidekick-style): taken powers appear as compact cards — power icon, name, level badge — with their slotted enhancements as an icon row (IO level under each), arranged in three level columns (1–12 / 14–28 / 30–49). Add-power choices moved below the build. Same clicks as before: click a slot to change it, right-click to clear.
- **Overview bar**: your build's vitals — typed and positional defense, resistance, recharge, HP, ST/AoE DPS — in one color-coded line that stays visible while you scroll. Green means you've hit the current-meta mark.
- **Tighter layout**: on wide screens the stats and controls panels sit side by side (three columns total), paddings and text are denser — much more of the build fits in one view.
- During a one-click update, the old browser tab now says plainly that the app reopens in a new tab and this one can be closed — and it refreshes itself into the new version the moment you look at it (browsers throttle background tabs, which is why it previously seemed stuck on the old version).

0.11.1 — July 3, 2026

- **The optimizer now completes winter sets to 6 pieces** when typed defense is the goal — the big fire/cold defense bonuses the master builder flagged as missed are now chased properly (typed-defense targets carry full priority in the solver).
- **Wizard-built kits include procs by default.** A freshly generated build has no slotting to "preserve", so the proc pass now always runs on it.

0.11.0 — July 3, 2026

The master-class release — the optimizer went to school on 4,000+ builds shared by one of the community's best builders, plus his direct coaching.

- **Damage procs are now priced.** The engine computes each slotted %Damage proc's real contribution (PPM math: recharge, cast time, area factor), so the optimizer can genuinely trade set

bonuses against procs — the core trade of the current meta.

- **Current-meta defense targets.** Presets now aim for ~35% Smashing/Lethal/Fire/Cold (typed) defense — or 35% Melee/Ranged/AoE for positional-armor characters — with the freed slots buying damage. Want the old 45% style? Put "classic softcap" in your goal. (Fire farm keeps its hard 45% fire floor — farms are still farms.)
- **Hasten gets 2 slots, never 3** — the third recharge IO is worth ~13% of face value after diminishing returns. A wasted slot, now unwasted.
- **One-click updates** — "Update now" downloads and installs the new version for you; the app closes, updates, and restarts itself. No more trips to the download page.
- Model v24; scoring is stamped, so older champion scores aren't compared to new ones.

0.10.1 — July 3, 2026

- **Fixed: power tray icons missing in the installed app.** Three data files (power icons, the proc catalog, incarnate magnitudes) were loaded with a path that only worked when running from source — in the packaged app they silently loaded empty. The tray now shows real power icons, and the solver's proc pass has its full catalog back.
- **The app finds your in-game saves for you.** After `/build_save_file` in game, click "Find my characters for me" — Hero Companion scans the usual Homecoming install locations, lists every character save it finds (newest first), and imports on click. Unusual install? Paste your game folder once and it's remembered. The manual file picker remains as a fallback.

0.10.0 — July 3, 2026

The installer release — Hero Companion now behaves like a proper Windows app.

- **Real installer**: download `HeroCompanion-Setup-0.10.0.exe`, run it, done. It installs to a standard location, offers a desktop icon, adds a Start Menu entry, registers in Add/Remove Programs with a working uninstaller, and upgrades in place. (The portable zip still exists for folder-preferring folks.)
- **No more console window**: the app runs in the background with a tray icon (the lime pulse, next to your clock). Right-click it to open the app or quit. Closing the browser tab no longer strands a mystery window.
- **Updates button in the header** — check for updates on demand, whatever you answered at startup.
- Your saves live in %APPDATA%\HeroCompanion as always — installing, upgrading, or uninstalling never touches your characters.

0.9.1 — July 3, 2026

- **Update prompts (opt-in)**: on first run the app asks once whether to check for updates at startup. Say yes and new releases show an "Update now / Remind me later" banner; say no and only the manual footer button ever checks.

- **Standalone builds are truly AI-free**: no more "AI: checking…" chip or "Ask Claude" box that could never work without a paid key. The panel is now the Build Assistant — presets, goals, and instant Solve, all fully offline. (Advanced: set HC_AI=1 with your own Claude key to re-enable the assistant.)
- **The app has its own icon** — the lime pulse badge, in the taskbar where it belongs, instead of the generic packaging icon.
- **Fixed**: the build wizard could clip its lower content on short windows with no way to scroll; it now scrolls within the dialog.
- **Fixed**: launching a second copy (or anything else holding port 5000) no longer kills the app — it moves to the next free port and says so.

0.9.0 — July 3, 2026

The "Hero Companion" release — the app has a name, a face, its papers in order, and a PC body to live in.

- **Standalone PC version**: Hero Companion now packages as a Windows executable — no Python, no setup; run it and the app opens in your browser. Your saves live in %APPDATA%\HeroCompanion.
- **Report a bug**: one click opens a pre-filled report (versions auto-included) on the project's GitHub page. Nothing is ever sent without your click.
- **Submit champion**: beat a shipped champion build? Export your candidate and post it to the submission queue — verified wins become the next shipped champion, with credit.
- **Check for updates**: compares your version against the latest GitHub release. Never automatic, never forced.
- **Hero/Villain reskin**: the app is now *Hero Companion* (or *Villain Companion* — flip alignment from the banner). Themed corners, colors, and cards; your choice is remembered.
- **Soldiers & Widows leveled honestly**: the level-1-to-50 walk now follows the real VEAT career — base powers only until the mandatory level-24 respec, then the full re-place order with your Crab/Bane or Night Widow/Fortunata branch.
- **Kheldians leveled honestly**: single-path sets auto-select, inherent flight means "no extra travel" by default, and the epic-pool step is gone (they don't get one).
- **Enhancement Converter planner + haul appraiser**: paste your drops (recipes included) and get keep/convert/sell verdicts with concrete converter paths and merit costs.
- **Origin pool rule enforced**: only one of Sorcery / Experimentation / Force of Will / Gadgetry / Utility Belt per build — the optimizer and validator both know the game won't allow a second.
- **Optimizer explores all pools**: Sorcery, Experimentation, and friends are now real candidates in the search, not just Fighting/Leadership/Speed/Leaping.
- **Terms, License, Credits**: CC BY-NC-SA 4.0 (free & noncommercial forever), Terms of Use, and full credits — including Guyver [SoV] of Sovereign, whose master builds were this tool's calibration standard.

- **This help system**: the Help button and this document.

Earlier (the road here)

- Physics/encounter model v23: wiki-verified to-hit, resistance, purple patch, mez protection, and archvillain rules; role-weighted scoring; scenario coverage.
- Deep optimizer: full-neighborhood search that runs to convergence with honest certificates, warm-started from champion builds, learning across runs.
- Leveling companion: per-level walk of the exact Homecoming schedule with cost-smart enhancement advice, deviation tracking, and end-game re-fit.
- Mids Reborn import/export compatibility.